

LEARNER GUIDE

Unit Standard Title: OPERATE A COMPUTER WORKSTATION IN A BUSINESS ENVIRONMENT
Unit Standard No: 114979
Unit Standard Credits: 2
NQF Level: 3

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This material has been compiled based on the **UNIT STANDARD 114979: Operate a computer workstation in a business environment**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by:

Training Dynamics Africa

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1. Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA

2. Purpose of this learning program

This Unit Standard is intended for learners who use a computer in a business environment. It is an introduction to basic computer literacy. The qualifying learner is capable of:

- ♦ Explaining basic IT concepts.
- ♦ Identifying and operating the hardware in a specific workstation.
- ♦ Using a computer operating system.
- ♦ Demonstrating basic maintenance procedures.

2.1 Target group

This program is compiled for the following **target group**:

- Security members
- South African Defense Force members
- South African Police Force members
- Correctional Services
- Individuals who wishes to complete the NQF level 3 National Certificate in security practices.

3. Standards and qualifications

It is important that you as the learner understand and realize that this training program is not presented in isolation, but that it was developed based upon nationally recognized standards known as unit standards.

Unit standards are the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

REMEMBER.....

Unit standards comprises of outcomes. An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.

4. Assessments

In order to assess whether a learner can actually demonstrate the desired outcomes, assessment criteria are included in the unit standard. Each outcome has its own set of assessment criteria.

REMEMBER.....

The assessment criteria describe the evidence that is needed that will show that the learner has demonstrated the outcome correctly.

Kindly refer to the unit standard attached hereto for the assessment criteria.

It is of utmost importance that the learner fully understands the assessment criteria as listed in the unit standard, as it is the only way in which the learner will know what he will be assessed against.

The **final or summative assessment** is the most important aspect of this training program. It is during this process that the learner will be declared competent or not yet competent.

The learner will know exactly how he will be assessed, and when and where he will be assessed. All of these details must be obtained from the training provider where the learner enrolled for his program.

5. Range statements

REMEMBER.....

Also included in the unit standard are the range statements in support of the assessment criteria.

The range statements indicate detailed requirements of the assessment criteria.

6. The learning guide

The learner guide is included in this material under various learning units. The learner guide has been designed in such a manner that the learner is guided in a logical way through the learning material and requirements of the unit standard.

On completion of this skill program, the learner will be assessed against the assessment criteria as stipulated in the unit standard.

Each of the modules consists of learner outcomes to be achieved during training and development interventions, work related experience or at your own time. The learner outcomes will enable you to achieve the required level of confidence and competence in order to undergo the summative (final evaluations) assessment.

The learner needs to assess his own knowledge and skill throughout the training process by completing the learner workbook. With the completion of the workbook you start collecting evidence to proof your competence. The learner workbook will be assessed (FORMATIVE ASSESSMENT)

TAKE NOTE

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

TAKE NOTE

- The best results will be obtained if you start with Study Unit 1 in the Learner Guide, and work your way through it, study unit by study unit.
- All learning outcomes are vital and must be studied (and exercised for all practical requirements) thoroughly to ensure that enough evidence for all specific outcomes is generated.

7. Learning assumed to be in place

It is assumed that a learner which registers for this program have completed a program in communication at NQF Level 3 or equivalent.

It is imperative to understand that this program are only one of the building blocks to complete Skills program 1 The complete list of unit standards for Skills program appears on the next page.

7.1 Security program matrix

SKILLS PROGRAM 1: SASSETA E				Patrol Security officer/Grade E
244177	Conduct a security patrol in area of responsibility	Level 3	7 Credits	
244184	Apply legal aspects in a security environment	Level 3	8 Credits	
244176	Use security equipment	Level 2	2 Credits	
244179	Handle complaints and problems	Level 3	6 Credits	
244181	Perform hand over and take over responsibilities	Level 3	2 Credits	
244182	Give evidence in court	Level 3	4 Credits	
12484	Perform basic fire fighting	Level 2	4 Credits	
116534	Carry out basic first aid treatment in the work-place	Level 3	2 Credits	
SKILLS PROGRAM 2: SASSETA D				Access control officer/Grade D
244189	Conduct access and egress control	Level 4	7 Credits	
242825	Conduct evacuations and emergency drills	Level 4	4 Credits	
11505	Identify, handle and defuse security related conflict	Level 4	12 Credits	
117705	Demonstrate knowledge of the Firearms Control Act 2000 (Act No. 60 of 2000)	Level 3	3 Credits	
113924	Apply basic business ethics in a work environment	Level 2	2 Credits	
119465	Write/present/sign texts for a range of communicative contexts	Level 3	5 Credits	
114979	Operate a computer workstation in a business environment	Level 3	2 Credits	
SKILLS PROGRAM 3: SASSETA C				Asset & reaction officer/Grade C
113909	Coach a team member in order to enhance individual performance in work environment	Level 3	5 Credits	
13912	Apply knowledge of self and team in order to develop a plan to enhance team performance	Level 3	5 Credits	
244578	Describe how to manage reactions arising from a traumatic event	Level 3	2 Credits	
113852	Apply occupational health, safety and environmental principles	Level 3	10 Credits	
13936	Outline the legal environment of a selected industry	Level 3	2 Credits	
119472	Accommodate audience and context needs in oral/signed communication	Level 3	5 Credits	
11508	Write security reports and statements	Level 4	10 Credits	

8. Preparation of summative assessment

After you have worked through the **LEARNER GUIDE** and you are satisfied that all theoretical and practical requirements/ evidence, as stipulated by the UNIT STANDARD, can be met, you need to make your own necessary arrangements with the assessor and/ or facilitator. The training establishment will then arrange a suitable date for you to attend your pre-assessment meeting and then arrange for a suitable date to complete the final assessment.

8.1 RPL assessment



The assessment of RPL learners will be conducted in the same way as for those of new learners. The assessment pack is exactly the same and will therefore be used for new learners as well as RPL Learners. It must however be noted that learners who are applying for RPL must provide proof of previous learning and subject related experience prior to the assessment.

This proof or evidence can be in the format of certified copies (certificates) of previous learning programs that have been attended.

All the evidence will be assessed and authenticated before a learner will be allowed to enroll for an RPL program.

9. Roles and responsibilities

9.1 Facilitator/instructor

- Prepare to facilitate.
- Transfer of knowledge and skills

9.2 The role of the assessor

- Plan for the assessment
- Prepare the learner for the assessment

9.3 The role of the Learner

- Prepare for assessment.
- Accept the responsibility of learning.
- Accept the responsibility to produce/deliver evidence.

STUDY UNIT 1

Basic understanding of Windows

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Demonstrate knowledge by explaining the basic meaning of a computer.
- ♦ Define the meaning of hardware.
- ♦ Demonstrate knowledge by identifying the following hardware components:
 - ⇒ Input devices
 - ⇒ Output devices
- ♦ Define the meaning of software.
- ♦ Demonstrate knowledge by explaining how software is operating within the operation system.
- ♦ Define application software.
- ♦ Demonstrate knowledge by explaining the different components of a Windows system.

1. What is a computer?

DEFINITION

A computer is a machine that takes in facts and information (known as data), and then processes (does something to or with) it. Afterwards it outputs, or displays, the results for you to see. Data consists of all kinds of information, including, pictures, letters, numbers, and sounds.

When talking of computers, there are two terms which must be correctly distinguished, hardware and software.

2. Hardware

DEFINITION

Hardware is all the parts of a computer that can be touched when the computer is taken to pieces.



- 1: Monitor
- 2: Hard Drive
- 3: Mouse
- 4: Keyboard
- 5: Scanner
- 6: Printer
- 7: Modem

2.1 Input and Output Devices

Input devices are simply the components that a person uses get data into a computer. The most basic input devices are a mouse and keyboard.

Keyboards



The Mouse:



Output Devices: Output devices are simply the devices that allow you to 'see' the data being produced by the computer. The most basic output device is the monitor. The printer is also an output device.



3. Software

DEFINITION

Software is all of the instructions that a computer uses to do what you ask it to. Pieces of software are often called programs and an operating system is a large program that helps all the other programs run.

There are two kinds of software:

- ♦ the Operating System (OS) and
- ♦ Applications Software.

3.1 Operating System

DEFINITION

Abbreviated OS, the Operating System is the piece of Software, which organises and controls the computer.

EXAMPLE:

. E.g. Windows Vista.

The Operating System runs the computer and the Applications Software. It makes sure that the Hardware and the Applications Software understand each other. This makes it the most important piece of software on the computer.

The Operating System also comes with utilities. These are pieces of Applications Software that mostly deal with managing data.

3.2 Applications Software

DEFINITION

Also called programs, pieces of applications software do the specific things you want.

Applications Software does the specific things you want the computer to do, Whereas the Operating System is general instructions to the computer for controlling the Hardware, Applications Software is specific instructions that work together with the Operating System to do work for you.

There are as many different kinds of Applications Software as there are things you could want to do with a computer, however the most common are:



- ◆ Word processing
- ◆ Spreadsheet
- ◆ Presentation, and
- ◆ Database software.

- ◆ Word processing software is used to write documents, which are formatted pages of text, such as letters or essays.
- ◆ Spreadsheet software organizes data, usually numbers, into columns and rows. It is used mostly for accounting and has many features for doing mathematical operations.
- ◆ Presentation software is used to make virtual (or, simulated) slide shows and usually have all sorts of exciting features for animations and sounds that you just cant do in a normal slide show.
- ◆ Database software is an advanced way of organizing complicated information in simple formats.

4 Windows and its Components

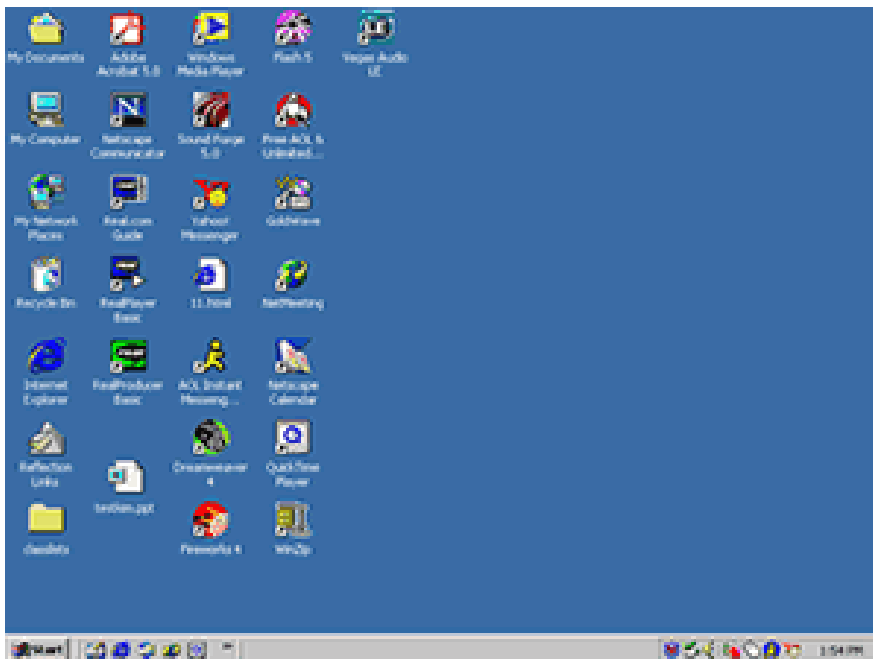
The main components of Windows when you start your computer are the:

- ◆ Desktop
- ◆ My Computer
- ◆ Recycle Bin
- ◆ Start Button
- ◆ Taskbar, and
- ◆ Shortcuts to applications.

After opening an application, you will also have a window with an active application which will appear in your desktop and taskbar.

4.1 Desktop

The Desktop is the very first screen you see after Windows starts. There you find the folders: My Documents, My Computer, the Recycle Bin and any Shortcuts for applications and files that you have created.



On the desktop are pictures with text labels under them, these pictures are called icons because they represent something else. If you move or get rid of an icon, all that means is that you have to access what they represent a different way, you haven't gotten rid of the application software the icon is representing. Icons usually represent programs, but sometimes they represent collections of data. Double-clicking (clicking twice in rapid succession) on one of these icons will open whatever it represents, the text tells you what it represents.

Icons: A picture that represents a program, data collection, or program function.

4.2 Taskbar

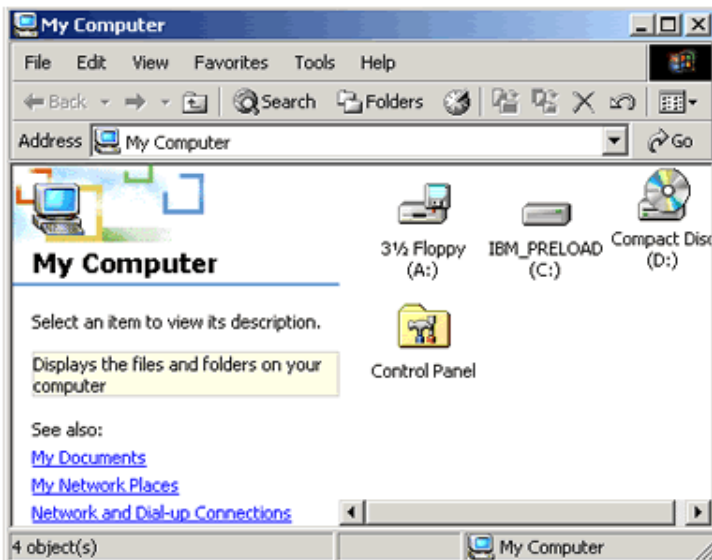
There is a bar that is usually at the bottom of the desktop, however it may also be on any other side. If you cannot see it, then move the cursor to the edge where it is and it will come up. It is called the task-bar.



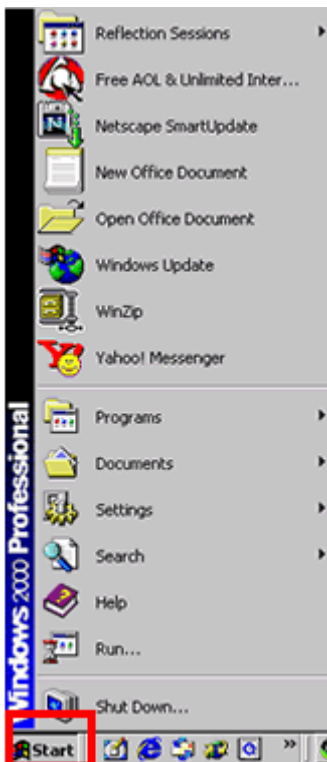
The taskbar is the bar along one side of the desktop. It is used for launching programs or opening the window of an open program.

4.2 My Computer

When you double-click on the My Computer icon on the desktop, it will open a window similar to the one shown below, where you can navigate between the computer's drives and Control Panel tools.



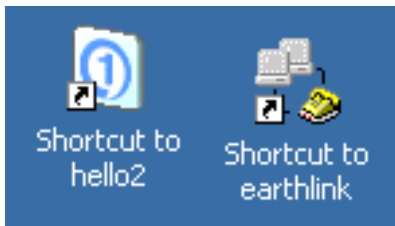
4.3 Start button



► The Start button is a button that opens the start-menu.

The icons that are on the desktop, the quick-launch bar, and the start-menu are usually shortcuts. On the desktop shortcuts are often indicated by a small symbol on top of the icon (↗). Shortcuts are what I meant earlier when I said if you move or get rid of an icon, all that means is that you have to access what they represent a different way, you haven't gotten rid of the actual thing. Icons represent all data collections and programs even if they are not shortcuts, however normally the ones on the desktop and in the start-menu are shortcuts.

4.4 Shortcuts to applications.



Shortcuts Icons that are only links to the things those icons represent.



The difference between icons and shortcuts is important so be sure you understand it. An icon is any picture that is meant to convey what something is. The icon on the start-button represents the fact that it is a major part of Windows, which is why it is a Windows logo. Shortcuts are a link to a program or data collection; the icon on a shortcut represents whatever the shortcut opens, however the same icon would be on the real thing as well.

[illegible]

STUDY UNIT 2

Basic skills

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Explain what a mouse is.
- ♦ Demonstrating the correct method of holding a mouse.
- ♦ Demonstrate the following mouse techniques:
 - ⇒ Left click
 - ⇒ Right click
 - ⇒ Click and drag
 - ⇒ Double click
- ♦ Explain the correct use of a keyboard.
- ♦ Demonstrate the correct method of using a keyboard by performing basic typing tasks.
- ♦

1. The mouse

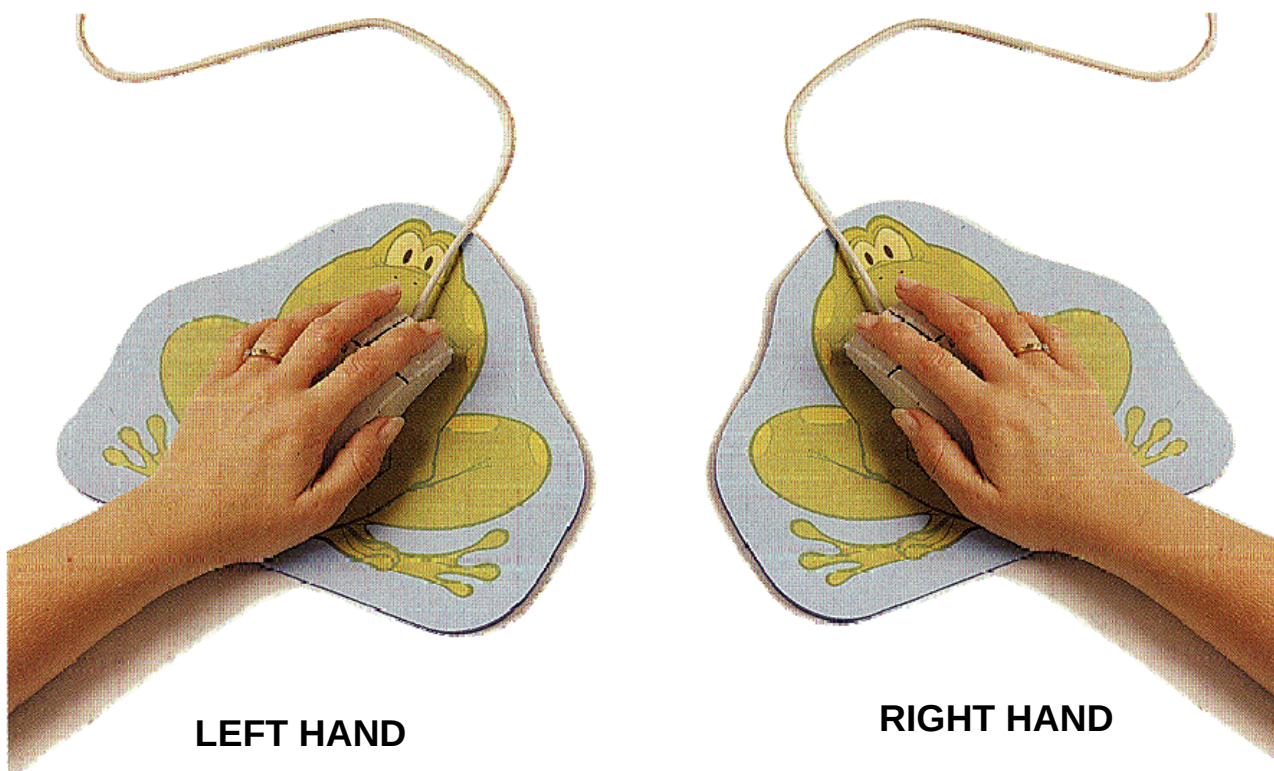
The mouse gives you control over your computer. With the mouse in your hand you cause the onscreen arrow to move around your computer display. It is based on the "point and click" philosophy.

You move your mouse to a point where you want something to happen, then you click to make that event take place. Sometimes you click once. Sometimes you click twice. And sometimes you hold down the mouse while your event is happening.

Most mouse devices give you two opportunities for clicking. There are both right and left areas to depress. The left area is generally used for issuing commands to your computer. To access special menus (known as context sensitive menus) use the right portion of your mouse.

1.1 How to hold the Mouse

"Let's shake hands." You know how to do that. Reach out and grab the mouse as if you were shaking hands. The rounded top fits into the palm of your hand and the cord extends away from you. Your thumb rests on the left side of the mouse, your index finger rests on the left button, your middle finger rests on the right button, and your remaining two fingers grip the right side of the mouse.



1.2 Definitions when using a mouse

- ♦ **Clicking:** Pressing down gently on the mouse button while holding the mouse perfectly still. The perfectly still part is important, but is not a problem if you press the button gently. New users sometime jerk the mouse as they click because they are trying to click too hard. Instructions which simply say "click," usually mean click with the left mouse button.
- ♦ **Left-Clicking :** Clicking with the index finger. Even if you are left-handed, if you have followed the instructions above and reversed the left and right buttons, you are using the index finger. So, left-click means index finger-click.
- ♦ **Right-Clicking:** Clicking with the middle finger. Even if you are left-handed, if you have followed the instructions above and reversed the left and right buttons, you are using the middle finger. So, right-click means middle finger-click.
- ♦ **Click and Drag:** Gently depress the mouse button and leave it depressed while you move the mouse. If you do this when the tip of your mouse pointer is over an object (title bar, icon, folder, etc.) your mouse pointer drags the object to a new location. When you let go of the mouse button, the object is dropped into a new location.

This is almost always done while holding down the left mouse button (index finger button), but there are some times when you may be instruct to click and drag using the right mouse button (middle finger button).

2. Keyboard skills

Mistakes happen quickly on the computer keyboard. Because of its sensitivity, the keyboard offers both challenges and rewards. Those rewards come after a bit of practice and study to take advantage of what the keyboard gives you. Mostly those advantages will be learned through practice and additional instruction. At this point we just touch some basic points. Here are some tips to make life simpler, and your work more efficient.

- ♦ Hold your hand with your thumb on the space bar, first finger on the "F" (left-hand) and "J" (right-hand). Then let your fingers naturally fall so that each rests on top of the next key along the same horizontal row.



- ♦ As you type, don't hold down the key after you depress it. If you do keep your finger depressed on a particular key, it will keep sending the computer commands. In this case it will be sending specific commands to repeat itself. So your result will be a string of unwanted characters displayed onscreen.



- ♦ Understanding the keyboard also means learning the special functions of certain keys. On the computer keyboard you often use two or more keys at the same time, indicated by the "+" sign. For example in managing applications you will learn that "Ctrl + Alt + Delete" is a decision of last resort. You depress these three keys simultaneously when you just want to get out of whatever muddle that the computer has left you with.



- ♦ You will see a key marked "Ctrl" This key helps you give the computer commands in combination with other keys (such as Ctrl+P to print a document in many programs). Holding down the shift key causes the keyboard to print either the upper case of an alphabetic character or the top character if two are displayed on the individual key.



[illegible]

STUDY UNIT 3

Basic IT concepts

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define a virus.
- ♦ Explain the consequences of viruses by providing at least three examples.
- ♦ Explain the importance of a company IT policy.
- ♦ Explain the consequences of an individual when not complying with a company IT policy.
- ♦ Define a desktop computer
- ♦ Define a network system
- ♦ Explain the difference between a standalone PC and a network.
- ♦ Explain the importance of security access of using a computer in your organization.
- ♦ Explain the concept of making backups.
- ♦ Explain the importance of making backups.
- ♦

1. Viruses

DEFINITION

A virus is a small piece of software that piggybacks on real programs. For example, a virus might attach itself to a program such as a spreadsheet program. Each time the spreadsheet program runs, the virus runs, too, and it has the chance to reproduce (by attaching to other programs) or wreak havoc.

EXAMPLE:

- ♦ **E-mail viruses:** An e-mail virus travels as an attachment to e-mail messages, and usually replicates itself by automatically mailing itself to dozens of people in the victim's e-mail address book. Some e-mail viruses don't even require a double-click -- they launch when you view the infected message in the preview pane of your e-mail software [source: Johnson].
- ♦ **Trojan horses:** A Trojan horse is simply a computer program. The program claims to do one thing (it may claim to be a game) but instead does damage when you run it (it may erase your hard disk). Trojan horses have no way to replicate automatically.
- ♦ **Worms:** A worm is a small piece of software that uses computer networks and security holes to replicate itself. A copy of the worm scans the network for another machine that has a specific security hole. It copies itself to the new machine using the security hole, and then starts replicating from there, as well.

2. IT Policy

It has become essential for organizations, no matter the size, to have an IT, or perhaps more exactly, a Computer Use Policy. Its purpose is two-fold: to protect the organisation and to protect the staff member.



REMEMBER.....

It must always be remembered that an organisation works within a community or society, and technology is one of the main ways of communicating with those outside the organisation. As such, technology is open to misuse which can do substantial business damage to an organisation.

Computer technology is also one of the most expensive investments an organisation has to make, and it is therefore important to protect the investment by ensuring correct operation of the equipment by staff.

But an IT Policy should also aim to have the protection of staff as one of its fundamental objectives. By clearly defining what staff can and can't do with company equipment, and ensuring that the use of computers is in line with an organization's overall ethos, staff can feel secure and comfortable in the knowledge they are operating within safe guidelines.

In addition an IT Policy should commit an organisation to take a responsible health and safety approach to the use of computer equipment, so that staff can carry out their jobs in a safe environment.

2.1 Non compliance on IT policy

Employees not complying with the company IT policy can suffer severe consequences. The consequences that could be suffered includes but are not limited to:

- ◆ Disciplinary action
- ◆ Blocking of internet and e-mail access
- ◆ Financial consequences

3. Desktop computer

DEFINITION

A desktop computer is a personal computer (PC) in a form intended for regular use at a single location, as opposed to a mobile laptop.

Prior to the wide spread of microprocessors a computer that could fit on a desk was considered remarkably small. Today the phrase usually indicates a particular style of computer case.

Desktop computers come in a variety of styles ranging from large vertical tower cases to small form factor models that can be tucked behind an LCD monitor. In this sense, the term 'desktop' refers specifically to a horizontally-oriented case, usually intended to have the display screen placed on top to save space on the desk top. Most modern desktop computers have separate screens and keyboards.



TAKE NOTE

The primary difference between a stand alone PC and a network is:

Standalone system is not connected to any other PC.

When two and more computers are connected with each other through a network cable or through a wireless system it becomes a network.

4. Computer network

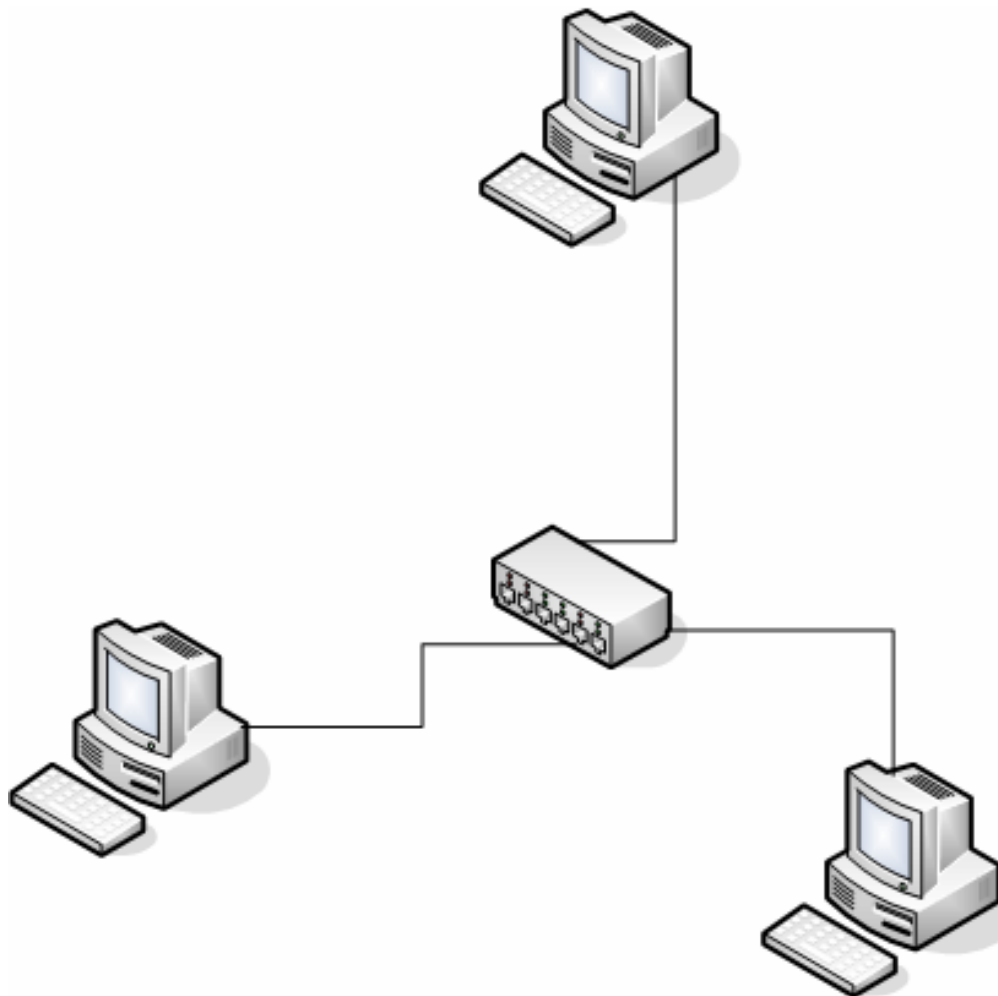
DEFINITION

A computer network is a system in which computers are connected to share information and resources. The connection can be done as peer-to-peer or client/server.

A computer network can be two computers connected:



A computer network can also consist of, and is usually made for, more than two computers:



4.1 Characteristics of a Computer Network

The primary purpose of a computer network is to share resources:

- ♦ You can play a CD music from one computer while sitting on another computer.
- ♦ You may have a computer with a CD writer or a backup system but the other computer. doesn't have it; In this case, you can burn CDs or make backups on a computer that has one of these but using data from a computer that doesn't have a CD writer or a backup system.
- ♦ You may have a computer that doesn't have a DVD player. In this case, you can place a movie DVD on the computer that has a DVD player, and then view the movie on a computer that lacks a DVD player.
- ♦ You can connect a printer (or a scanner, or a fax machine) to one computer and let other computers of the network print (or scan, or fax) to that printer (or scanner, or fax machine)
- ♦ You can place a CD with pictures on one computer and let other computers access those pictures.
- ♦ You can create files and store them in one computer, then access those files from the other computer(s) connected to it.

5. Security access

Windows is an operating system which manages sessions, meaning that when the system is started, it is necessary to log in with a user name and password. When Windows is installed, the administrator account is created by default, as is an account labeled guest.



It is possible (and recommended) to modify user permissions (which actions they have a right to perform) as well as to add users with the user manager.

A user account is an identification uniquely assigned to user to allow him or her:

- ♦ to log on to a domain in order to access network resources,
- ♦ to log on to a local computer in order to access local resources Therefore, each user who regularly uses the network must have an account.

5.1 Managing users

The user manager is the standard utility provided with Windows NT, which manages users (as its name would suggest). It is available in the Start User management in Windows.

5.2 User accounts and security

There are two kinds of accounts in Windows.

Built-in accounts are accounts that you create. After installation, Windows is set up with built-in accounts:

- ♦ the default accounts administrator
- ♦ and guest, which provides only minimal security.

The different accounts are:

- ♦ **Accounts you create:** User accounts for logging onto a network and accessing network resources. These accounts contain information on the user, in particular his or her name and password.
- ♦ **Guest:** This lets sometime users log on and access the local computer. By default, it is deactivated.
- ♦ **Administrator:** Used for managing global configuration of computers and domains. This account can carry out any task.

IMPORTANT

It is essential:

- ♦ first, to deactivate the guest account which would let any user log in to the system.
- ♦ and second, to change the name of the administrator account in order to reduce the risk of intrusion by that user account. As the administrator account has full permissions, it makes a prime target for would-be intruders.

TAKE NOTE

Organizations will create their own accounts through the network administrator to allow access to information to specific employees. Not all employees will have access to all information.

The company policy can be consulted to determine your access.

6. Backups

6.1 The Importance Of Backups

Backups are data insurance. Real-world insurance policies protect your home, car, and health from financial disaster. Likewise, data backups protect the work you do on your computer. In a way, having an exact copy of the files on your computer stored elsewhere is even better than insurance. Insurance only kicks in after a tragedy. If your hard drive breaks down, falls victim to a virus, or deletes some important files in response to your careless commands, it's no tragedy when you have a backup. And if the backup is recent and serves as an exact copy for all the data on your computer, the recovery process shouldn't present any problems or delays at all.

Not having a backup is a frightening prospect. If the worst happens, how much data will you lose? How much time will it take for you to re-create data that has not been backed up? With a proper backup routine, you can recover your data from the clutches of a full system failure. Plus, you can recover your OS (operating system), applications, documents, e-mail, and other settings.

It is important to back-up for the following reasons:

Mechanical or electronic failure could mean that you lose important data. Re-inputting the data is costly to the establishment, both in time taken to do it and in delays in providing services and / or products. Furthermore the loss of information could also inconvenience clients in:

- ♦ delays in billing, incorrect billing, etc, which could result in loss of revenue.

6.2 Types of back-up systems

You can back-up onto the following:

- ♦ **Magnetic tape:** This is considered reliable if properly cared for and can hold a large amount of data.
- ♦ **Exchangeable magnetic disks:** These disks range in size and have very high data transfer rates. They are vulnerable to accidental damage and dust and therefore need to be handled with care.
- ♦ **Removable cartridge disks:** These are sealed disks that require an additional drive in order to run.
- ♦ **Floppy disks:** These are used for immediate copying and longer-term back up of data. However their capacity is low and speed of access is slow. Floppy disks can also become corrupt and data can be lost in this way.
- ♦ **Videotape:** This has not been exploited as yet. They are able to hold large amounts of data and are relatively cheap.
- ♦ **Optical disks:** These are capable of holding large amounts of data. They are often referred to as CD-ROM or CD's and can be written to by users. Two types of CD's are available: the one you can copy files to only once and the other is a re-writable CD, which means you can erase old data and copy new data onto the CD.

6.3 Establish A Backup Policy.

How often, to what media (plural term for computer storage material such as diskettes, CD-Rs [CD-recordable discs], and tapes), and which files you back up depends on your specific needs. Not everyone needs to back up his or her files on a daily basis, or to the same media type. Also, other users may not need to back up the same type of files that you do. In this section, we'll explore all these issues.

6.4 Sample of backup policy and procedures.



It is important to check out your company policy and procedures for backup policies and procedures.

[illegible]

STUDY UNIT 4

The hardware on your PC

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Identify the components on a PC.
- ♦ Use the correct cables to connect the components onto the PC.
- ♦ Demonstrate the skills to turn your PC on.
- ♦ Demonstrate skills by turning your PC off in the correct way.
- ♦ Demonstrate the correct method to log on onto your PC by using a mouse.
- ♦

1. The "Computer"

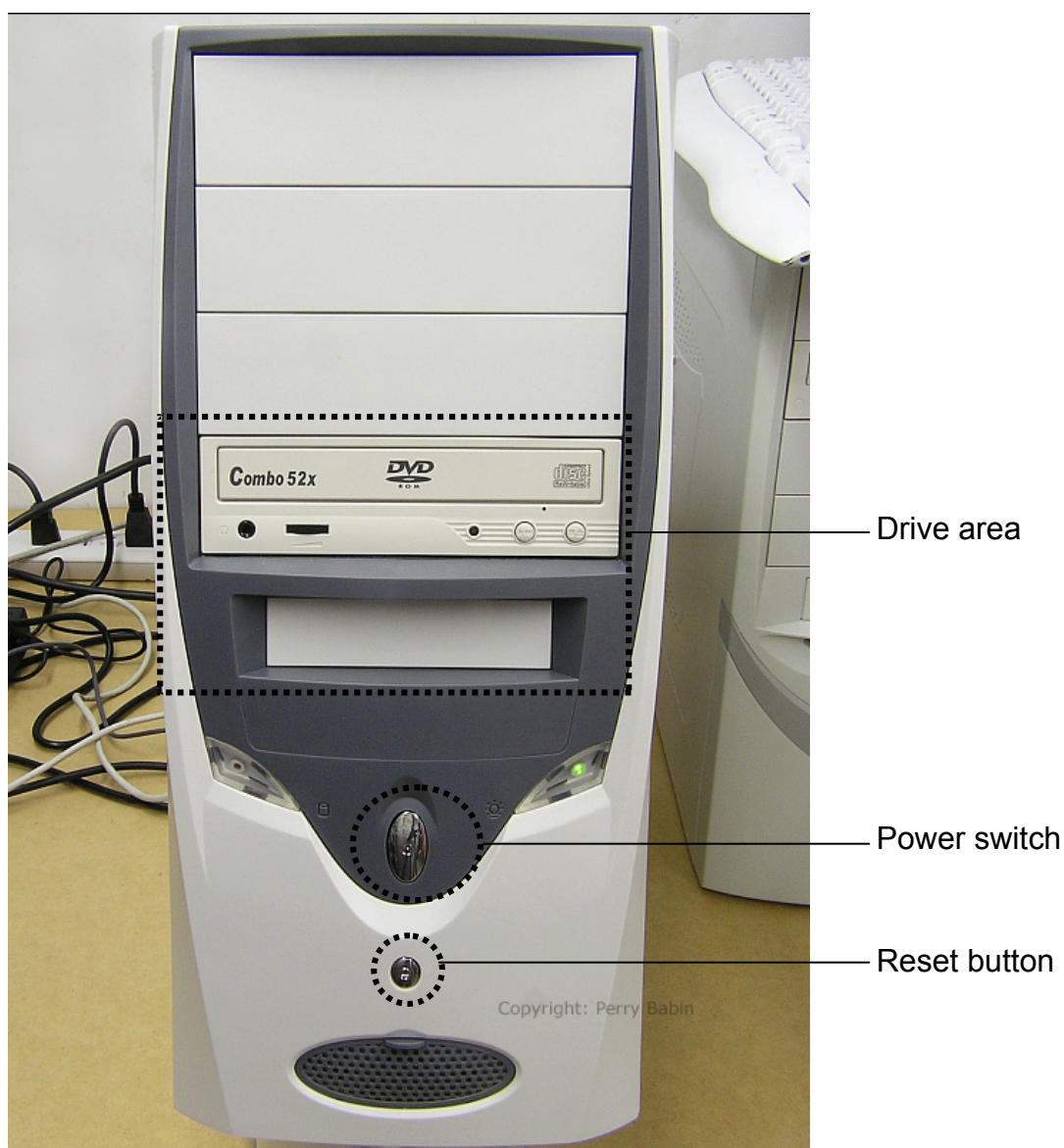
The part of the computer that most people refer to as 'the computer' is simply a box that holds most of the components that make up a personal computer (a 'PC').

2. Front Panel Buttons/Switches

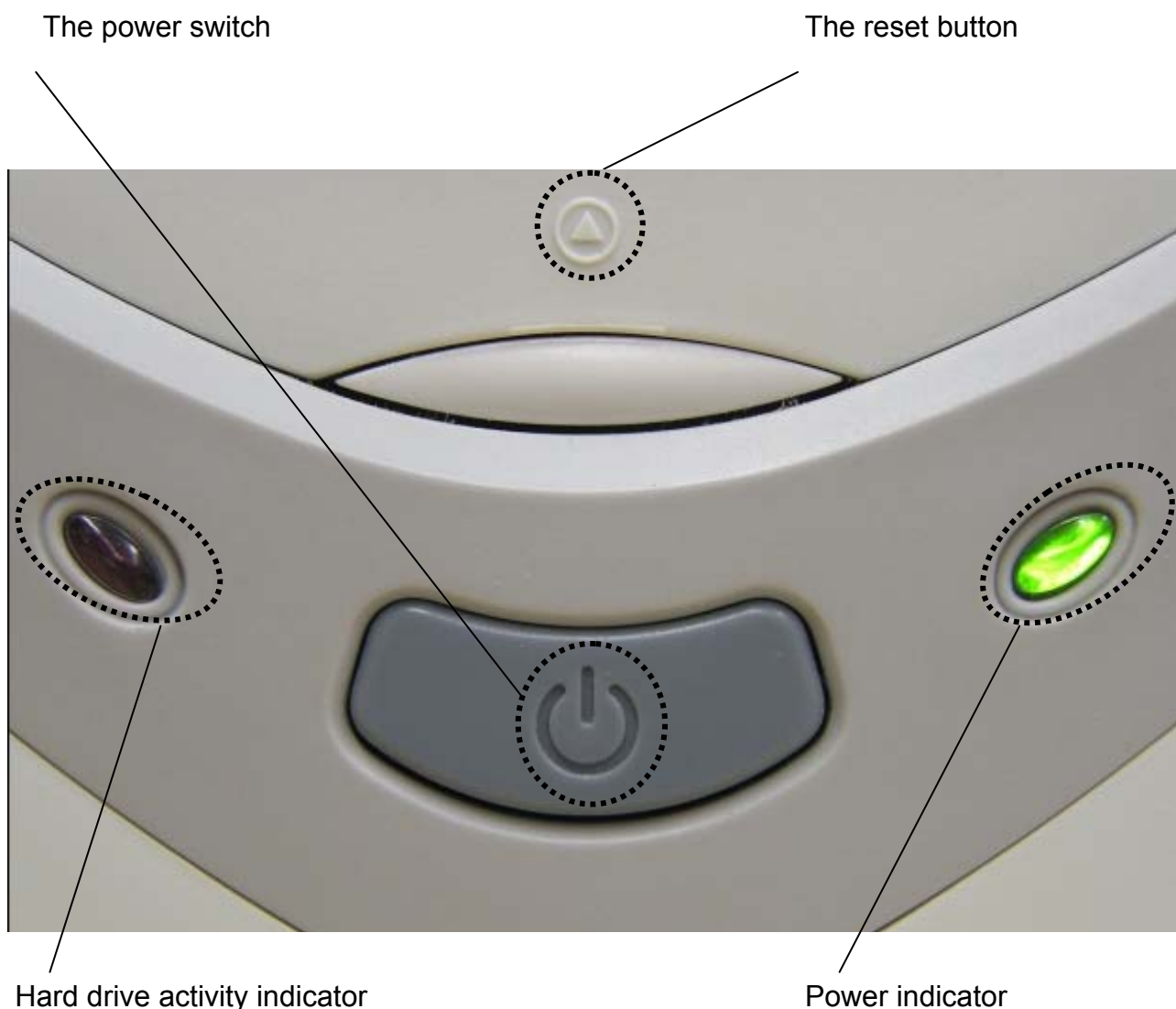
In the following picture, you can see the front of a basic computer. It has only one optical drive (a drive that uses a laser to read the data from the media) and no floppy drive (many newer computers do not have floppy drives).

Below the drive area, you can see that there is a large oval button. That's the power switch. Its used to turn the computer on and off.

The smaller button below the power button is the reset button (only to be used in the case of a serious crash where the computer freezes).



I mentioned the 'power' and 'reset' switches. On new computers, they have somewhat universal markings to indicate which is power and which is reset. In the following image, you can see the front of a computer. (a different computer)



3. Front Panel Lights/Indicators

To the left and right of the power switch, you can see two LED indicators. The left indicator is the hard drive activity indicator. It is lit when the computer is accessing the hard drive. In the picture it is not lit and you can only see a hint of red.

On the right, you see the green 'power' indicator. It's lit whenever the computer is on. On this computer, the colors red and green are used but the colors for the hard drive indicator and the power indicator will vary from one manufacturer to another. Don't let it confuse you.

4. Rear Panel Switches

The back of the computer can look intimidating to computer newbies but it's really pretty simple.

At the top of this computer, you can see where the power source (from the AC mains - the household wall outlet) plugs in. It is plugged into the 'power supply'. The power supply converts the mains voltage 115VAC/230VAC to the various voltages needed by the computer

On this particular power supply, you can see 2 switches. One is the main power switch. The other is the voltage selector switch. The main power switch completely cuts power to the power supply. Switching this off is essentially the same as unplugging the cord from the power supply. The voltage selector switch allows the power supply to operate with different mains voltages.



5. Rear Panel Connectors



On the rear panel of any modern computer, you'll find several common connectors. This is a quick list...

- ♦ At the top of the connector panel (just below the power supply), you will see two connectors. One has a purple plug in it. The other is open. These are PS/2 ports and are used for the keyboard and mouse. Typically, they are color-coded to match the connectors.
- ♦ Just below the PS/2 ports, you see a 9 pin d-sub connector. This connector (with 9 'pins' instead of tiny sockets as are found on the other d-sub ports) is a serial port. It's used for data transfer.

6. Turning it on

- ♦ If it's a recent Windows computer it'll have a button on the front with the Universal Symbol for power.
- ♦ If it's an older model, look on the front for a power button of some kind

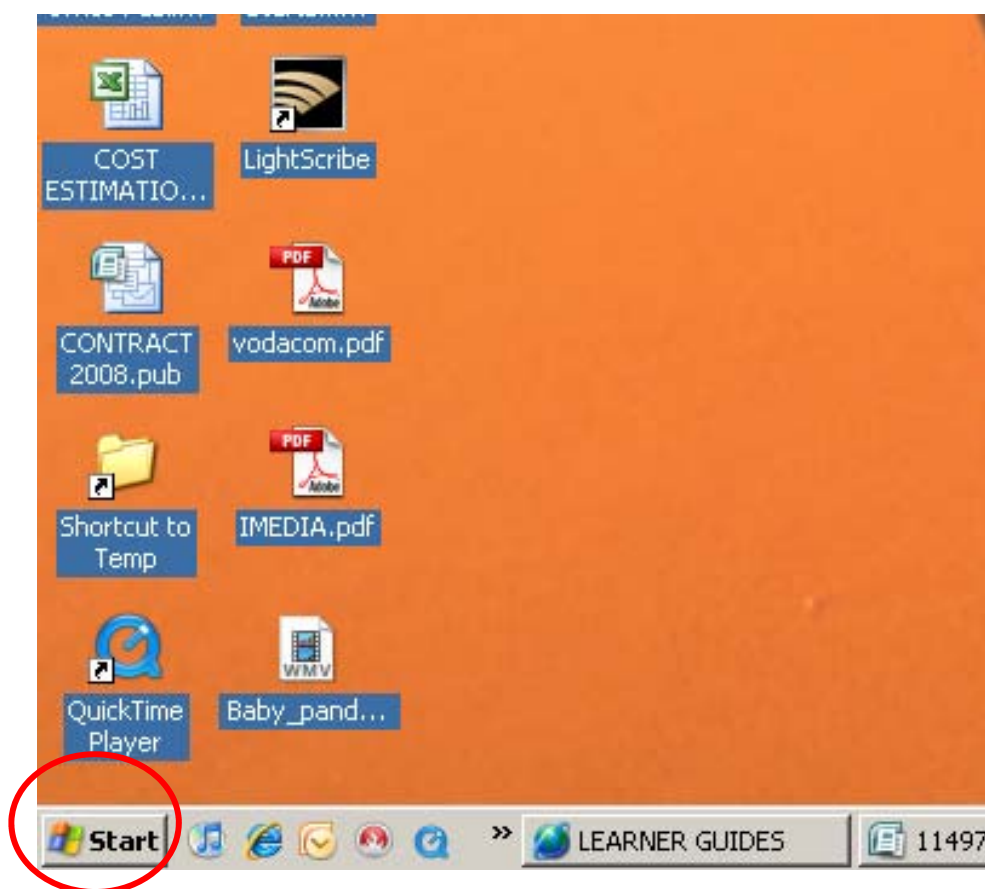


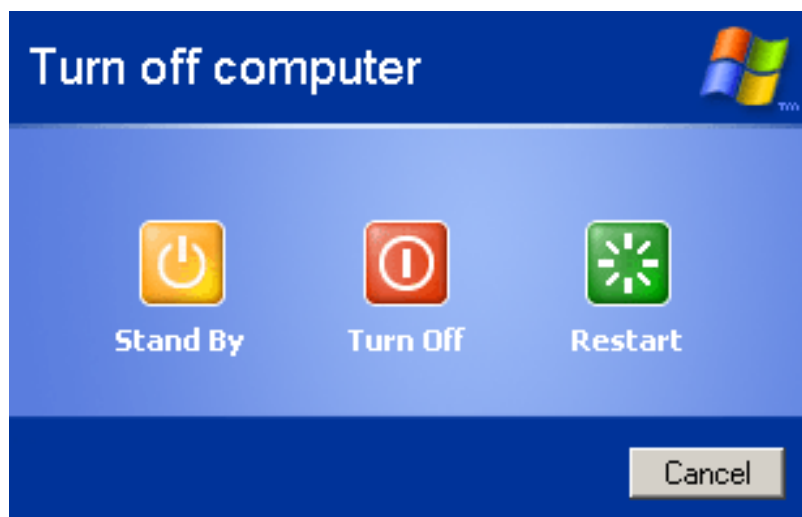
The Universal Symbol for On-Off Power Switches

7. Turning it off

Do not just turn the machine off .

- ♦ Unless it's crashed and then you have to,
- ♦ Use the Start button to go to Shutdown, click Shutdown and then OK.
- ♦ Then wait. If it's a recent PC it will power itself off. If it's an older model you'll see a message saying it's OK to turn off the power.





8. The login box

Once you have WAITED until Windows comes up (a.k.a.: Boots) you get a dialog box asking you to log in. It looks like this:



If you're using Windows v95 or v98 this dialog is a lie. Just click Cancel and it goes away!

TAKE NOTE

Note, however, that any version of Windows can be set to require a login. If you see this and click Cancel and it does not go away, contact your systems administrator or technical support staff. You might have a Windows computer with the security turned on. They'll have to give you a login and password that will work for your Windows computer.

Note also that Macintosh computers may have security (either added in or as part of the system). But on Macs you have to take a step to make it show up. It is not, by default, on and meaningless.

[illegible]

STUDY UNIT 5

Use the computer

LEARNING OUTCOMES

On completion of this study unit the student will....



- ◆ Demonstrate basic skills by providing a demonstration based on a specific task on the following:
 - ⇒ Highlight texts
 - ⇒ Copy/Cut and Paste Text
 - ⇒ Save and/or Save As
 - ⇒ Print Command
 - ⇒ Create New Folder
 - ⇒ Copy/Move Files
 - ⇒ Drag and Drop
 - ⇒ Create a shortcut
 - ⇒ Rename/Delete a Folder or File
- ◆ Explain what a file extension is.
- ◆ Explain the concepts of:
 - ⇒ A Hard drive
 - ⇒ A floppy drive
- ◆ Explain the use of the recycle bin
- ◆ Identify basic software programs and explain their basic use
- ◆ Demonstrate the use of the help function on a PC.

1. Highlight texts

Put your cursor at the beginning of a word or sentence you want to highlight, then click and drag your mouse to the end of the selection.

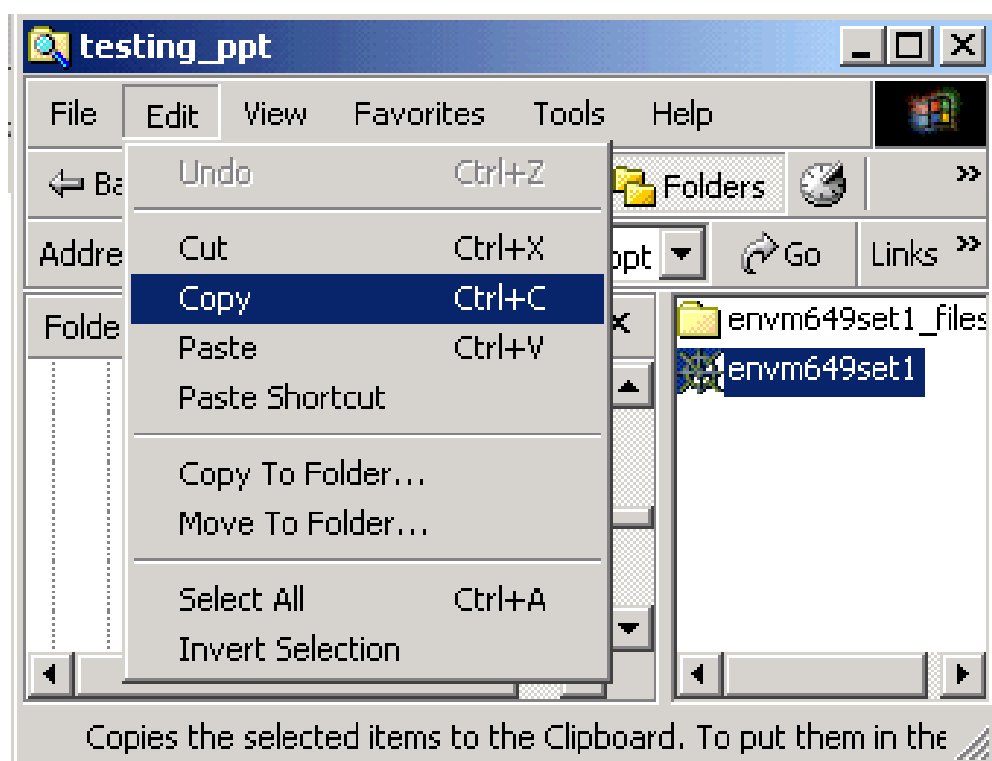
Basic Computer Skills
for a Windows Environment

Basic Computer Skills
for a Windows Environment

2. Copy/Cut and Paste Text

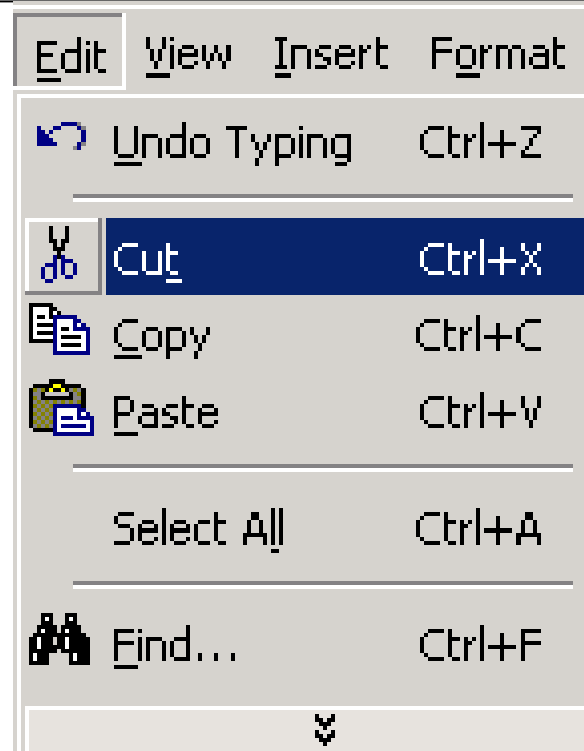
Copy is used when you want to have an area of text from a document twice at the same document or in another document.

You first select the area of the text by highlighting it and then press CTRL+C, or go to Edit menu and select Copy.



Cut is used to delete an area of text from a document.

You first select the area of the text by highlighting it and then press CTRL+X, or go to Edit menu and select Cut.



Paste is used to insert an area of text or image in a document by hitting CTRL+V or in the Edit menu selecting Paste.

Another way of using the copy, cut or paste commands is using the right-click button of your mouse.

- ♦ After highlighting the text you right-click and a menu will appear where you can select Copy or Cut.
- ♦ To paste you just insert your cursor where you want, right-click and select Paste.

2.1 Copying or cutting and pasting a text in Microsoft Word

- ♦ Highlight the text you want to copy or cut, press CTRL+C (for copy) or CTRL+X (for paste).

Can you answer the questions below?

Do you understand the concepts and computer terminology?

If not, be sure to sign up for a Basic Skills technical workshop before undertaking the WebTycho training.

- ♦ Insert your cursor where you want to paste the text.

Do you understand the concepts and computer terminology?

If not, be sure to sign up for a Basic Skills technical workshop before undertaking the WebTycho training.

- ♦ Paste your text by pressing CTRL+V. The text will appear.

Do you understand the concepts and computer terminology?

Can you answer the questions below?

If not, be sure to sign up for a Basic Skills technical workshop before undertaking the WebTycho training.

3. Standard and Formatting Toolbars

Toolbars are rows of buttons and boxes. When you click on a button or box, an application opens or a task is performed. Toolbars normally appear at the top of an application's window.

- ♦ **The Standard Toolbar** shown in the image below is from Windows Explorer; the standard toolbars in other applications may have buttons that allow you to open a new document, save, print, and spell check.



- ♦ **The Formatting Toolbar** is used to format a text using alignments, font type, font size, bold, italic, and lists. It is provided in most applications. The image below shows the Formatting Toolbar from Netscape Composer.



4. Save and/or Save As

Save is used when:

- ♦ It is the first time you are saving a file.
- ♦ You want to keep saving an existing file in different moments that you are working on it.
- ♦ It is the end of a working period.

Save As is used when:

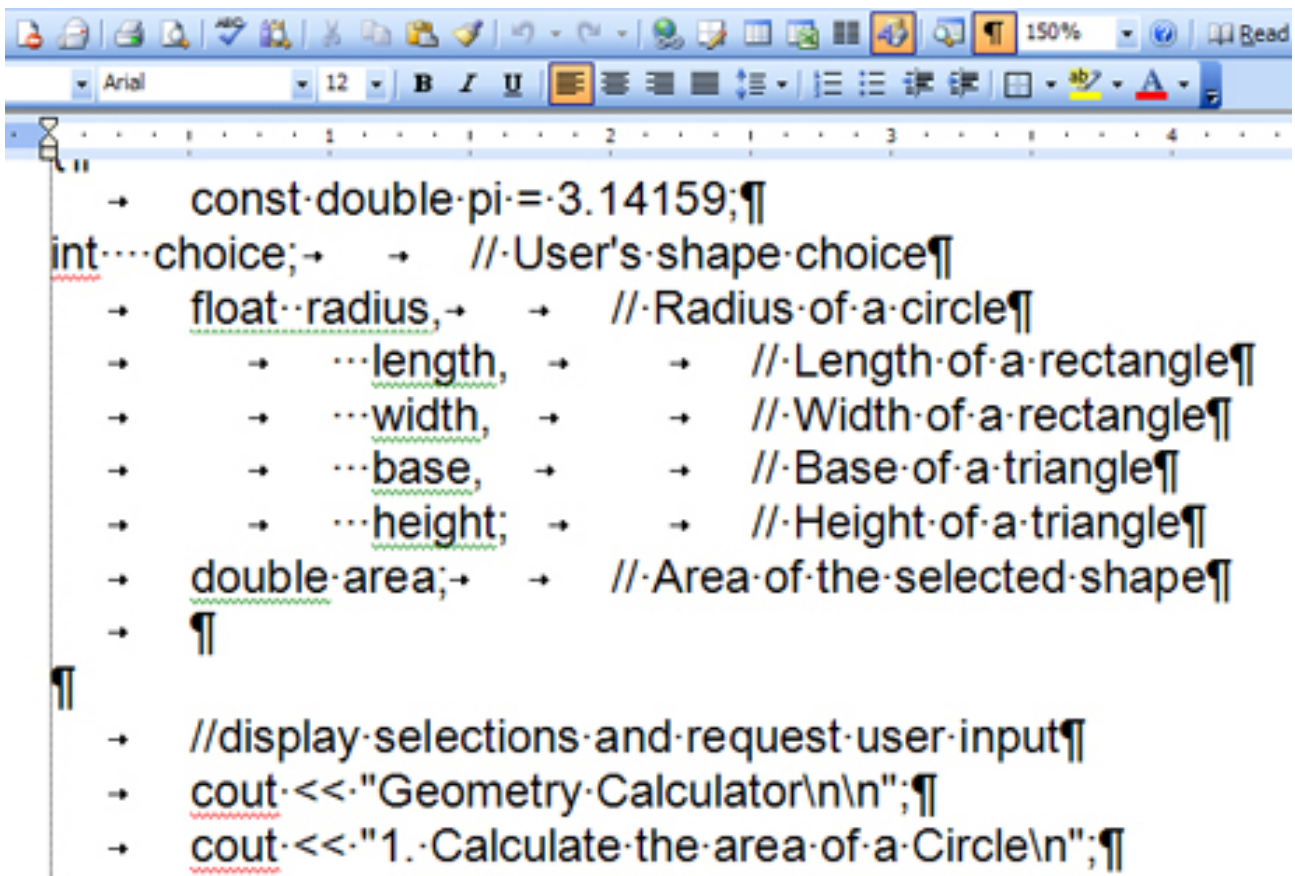
- ♦ You want to save a file with a different name to create backups of a file in the creation process. In that way, you can have different versions of the same document and can return to a previous stage.

When you first create a file, it is saved in temporary memory. Should your computer lose power or should you turn off your computer, the file will disappear.

To actually save the file you have to transfer it from temporary to permanent memory. That involves putting the file on some sort of storage medium like a hard drive, USB drive, floppy disk, or CD-ROM.

The actual process is straightforward.

- ♦ Have your file open and onscreen. Do not highlight any part of it and don't worry that not all of the file actually appears.

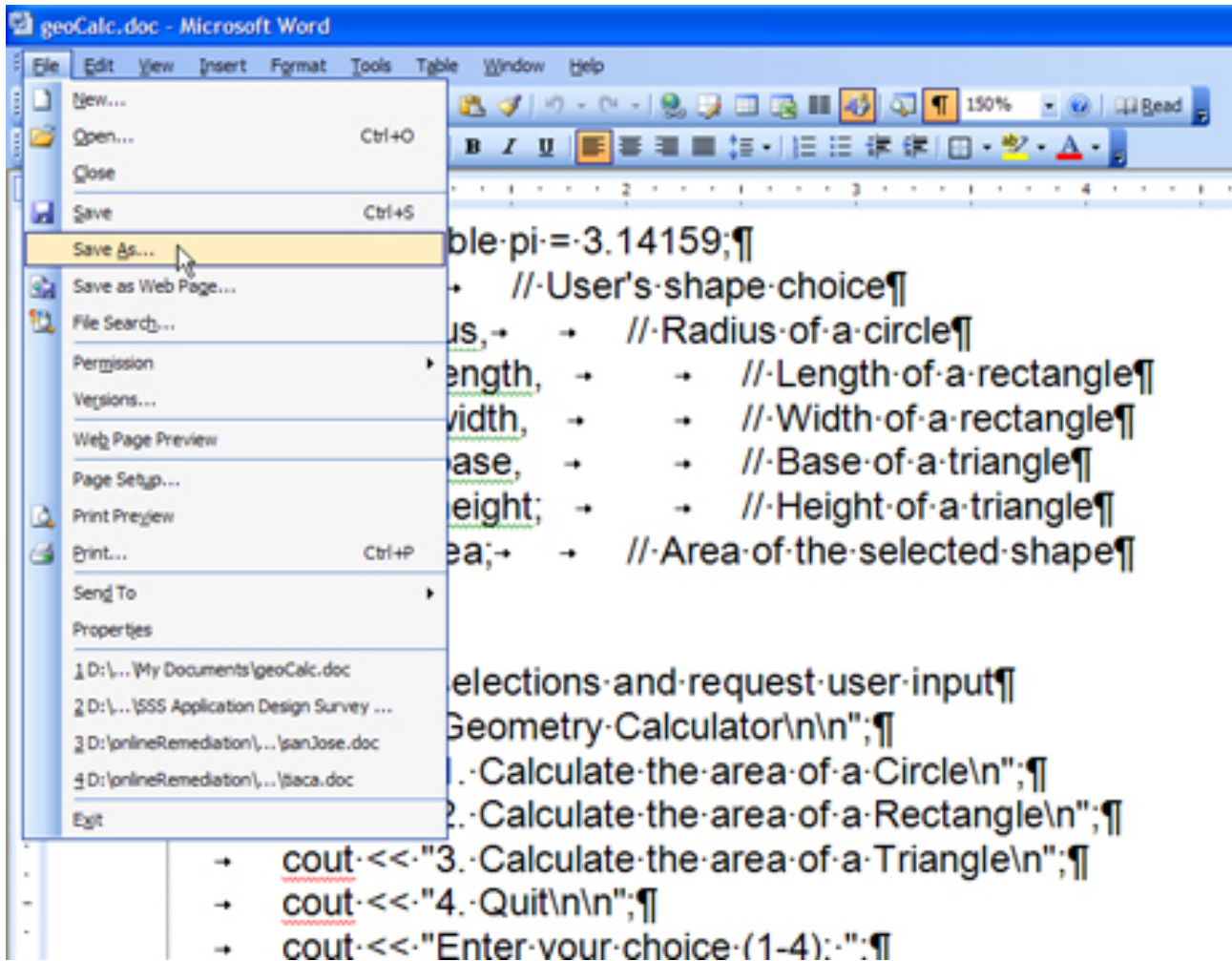


```
→ const double pi = 3.14159;
int choice; // User's shape choice
float radius; // Radius of a circle
→ length; // Length of a rectangle
→ width; // Width of a rectangle
→ base; // Base of a triangle
→ height; // Height of a triangle
double area; // Area of the selected shape
→

→ //display selections and request user input
→ cout << "Geometry Calculator\n\n";
→ cout << "1. Calculate the area of a Circle\n";
```

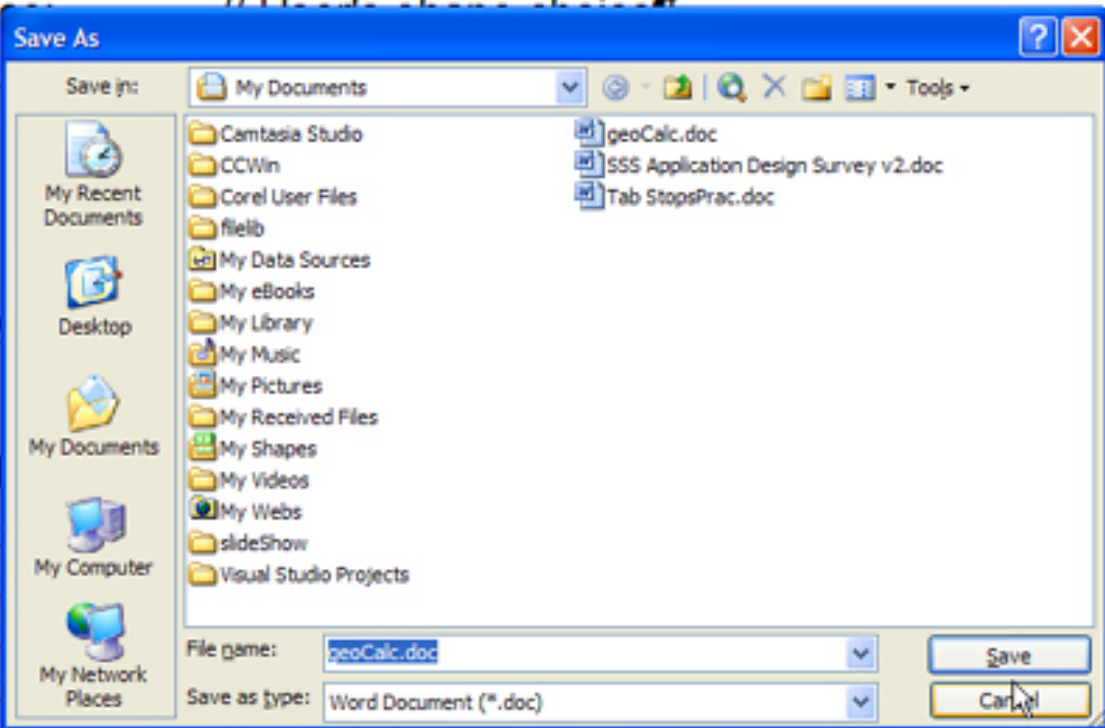
- ♦ When you save a document for the first time, you will need to supply a name for that document.

To do that go to File > Save As. The File dropdown menu appears. When you click Save As, an onscreen box will appear onscreen asking you to make some choices.

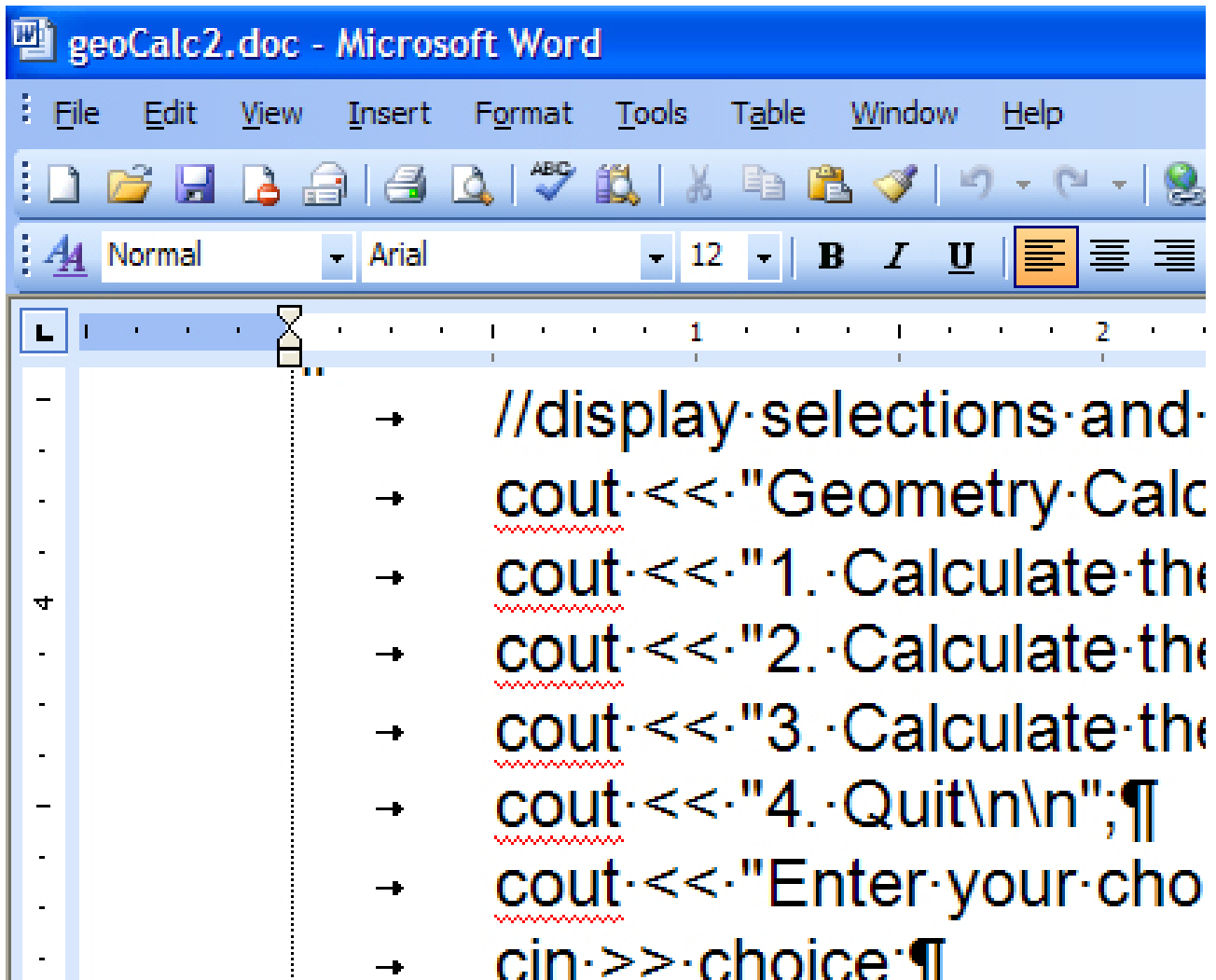


Near the bottom of the box you should fill in a name where there is a blank next to File Name. When deciding on a name, try to make it have something to do with the file's actual content. Also, try to be consistent in how you name files. This will help you remember the names when you want to open that saved file.

Notice that the file at right is being saved to "My Documents". (under the blue bar.) This is a convenient place to store your files. Another convenient place to locate these files is on the Desktop.

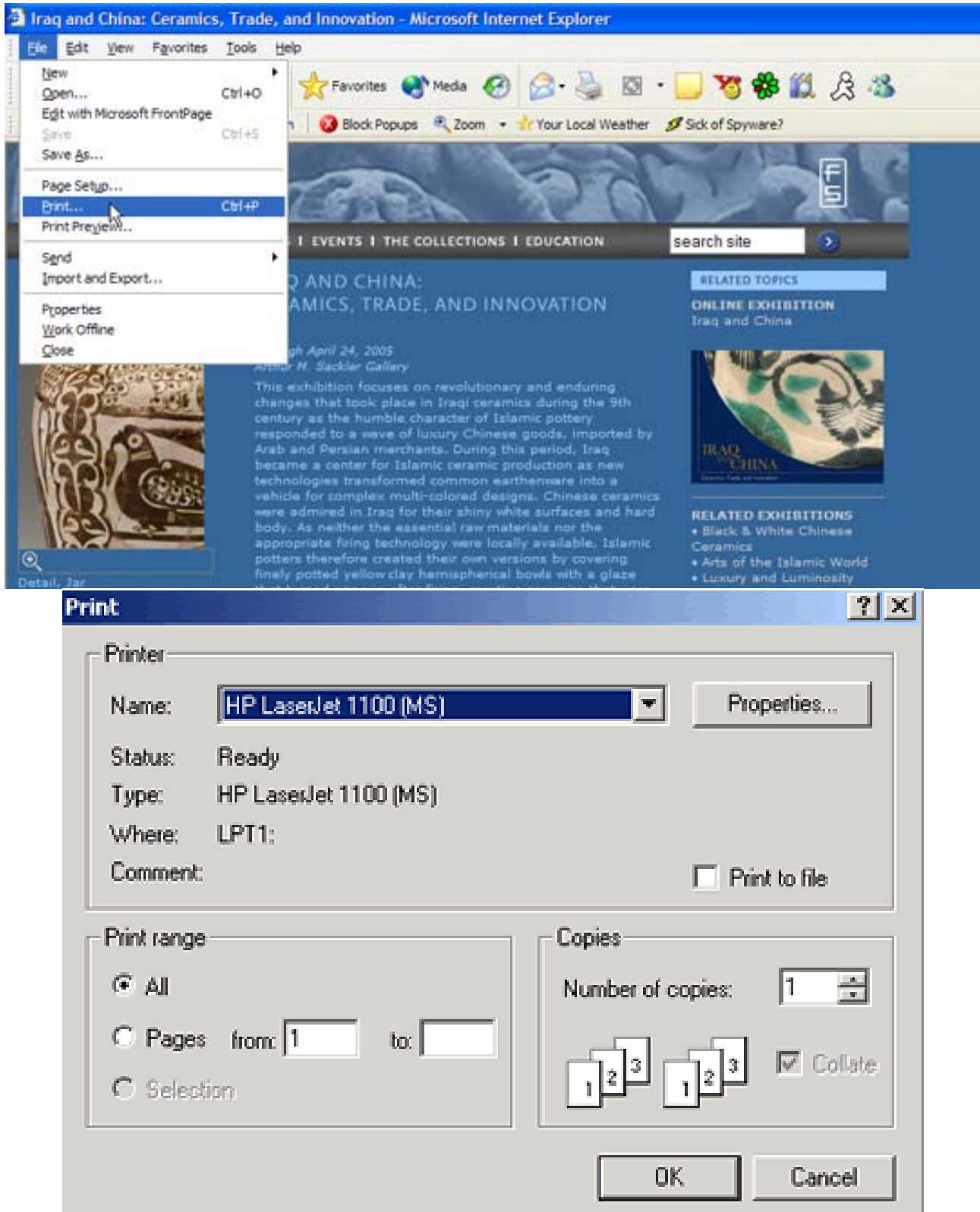


After you have named the file, that name will appear in the blue bar at the top of your screen. If it does not appear, then you have not saved properly. Just repeat the process in this case. You only need to do a Save As one time for any document. However, you can do it a second time to save a copy of the document under a separate name. Once the Save As process is complete, simply do File > Save or use the keyboard shortcut Ctrl + S to save your files as you work.



5. Print Command

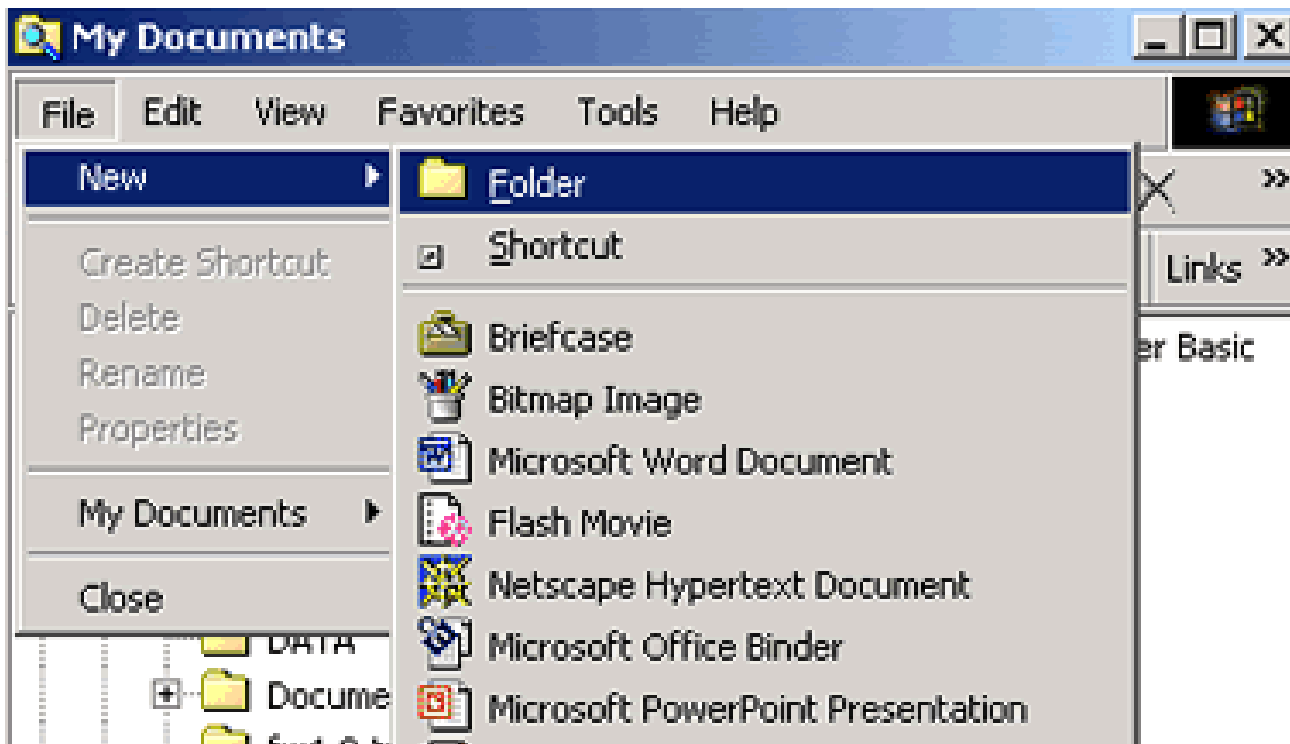
To print a page go to the File menu and select Print. A window will pop up with options for how many copies and which pages you want to print. Other options are also available depending on the printer.



6. New Folder

To create a New Folder, which means a new directory (a place where documents from the same subject are stored), go to:

- ♦ the File menu,
- ♦ then New and
- ♦ Select Folder from the pop-up menu, either in My Computer or in Windows Explorer.
- ♦ To open Windows Explorer, click the Start button, select Programs/Accessories/Windows Explorer.



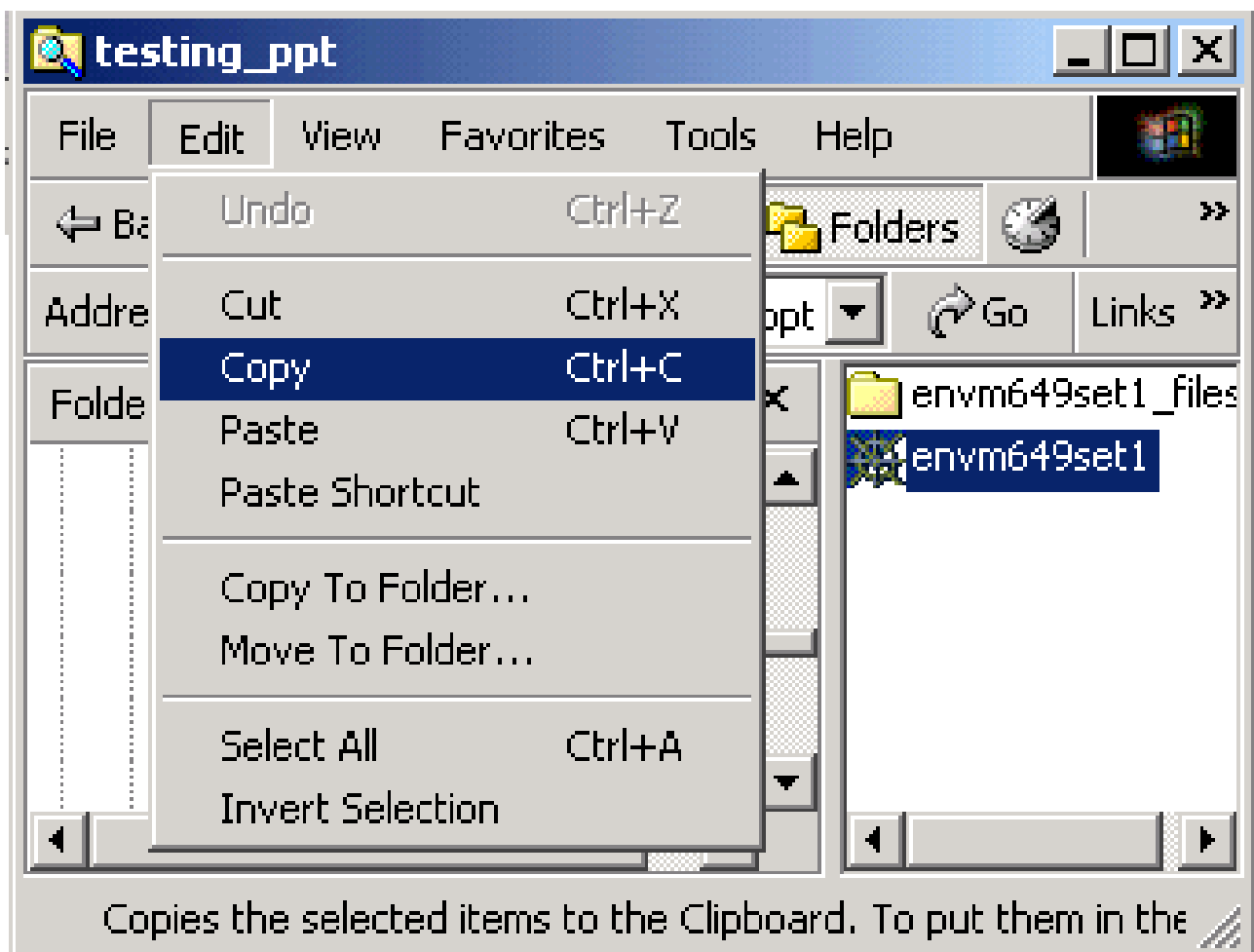
7. Copy/Move Files

To copy and move a file or folder, use Windows Explorer.

To open the Explorer application, click the Start button and select Programs and Windows Explorer (some computer have Explorer under Programs/Accessories).

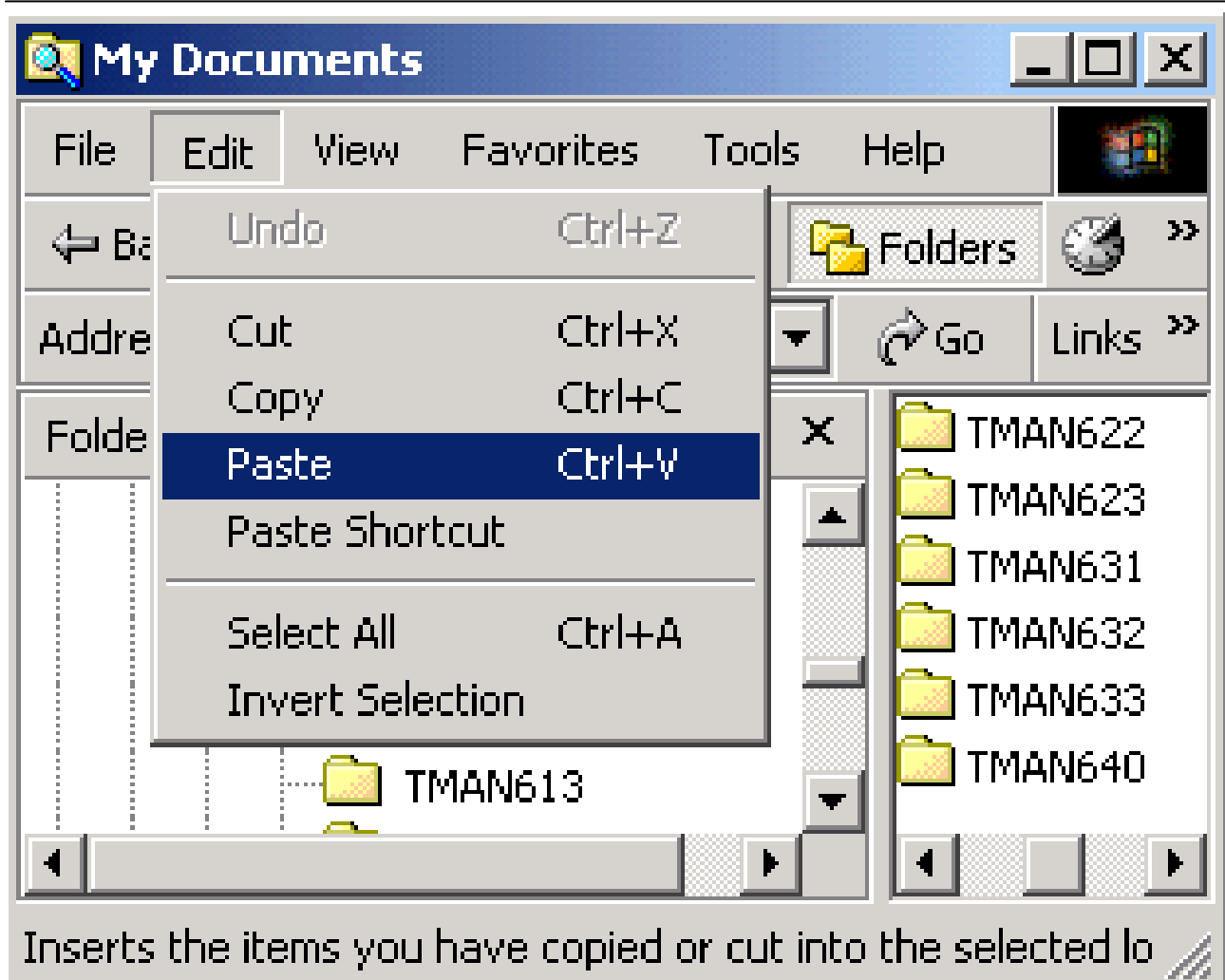
Copy or move a file using Menu command:

- ♦ Select the disk drive that contains the file by clicking twice on it.
- ♦ Click on the folder where the file is located
- ♦ Select the file by clicking on it.
- ♦ Go to Edit menu, select Copy (if you want to retain a copy of the file in the original place) or Cut (if you want to move the file).
- ♦ Navigate to the new location where you want to paste the file.
- ♦ Go to Edit menu, select Paste.



TAKE NOTE

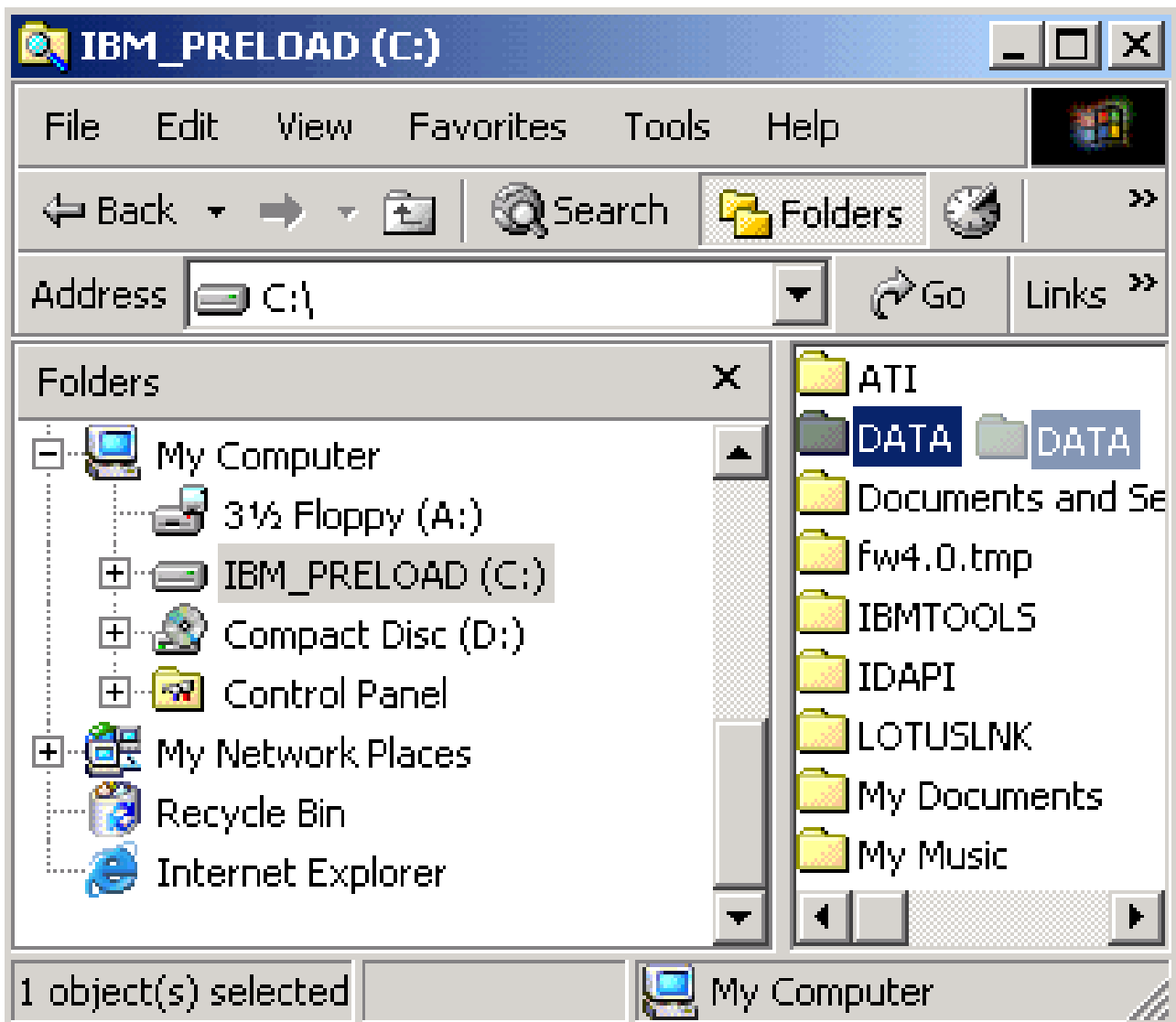
The same procedure can be used if you want to copy or move an entire folder.



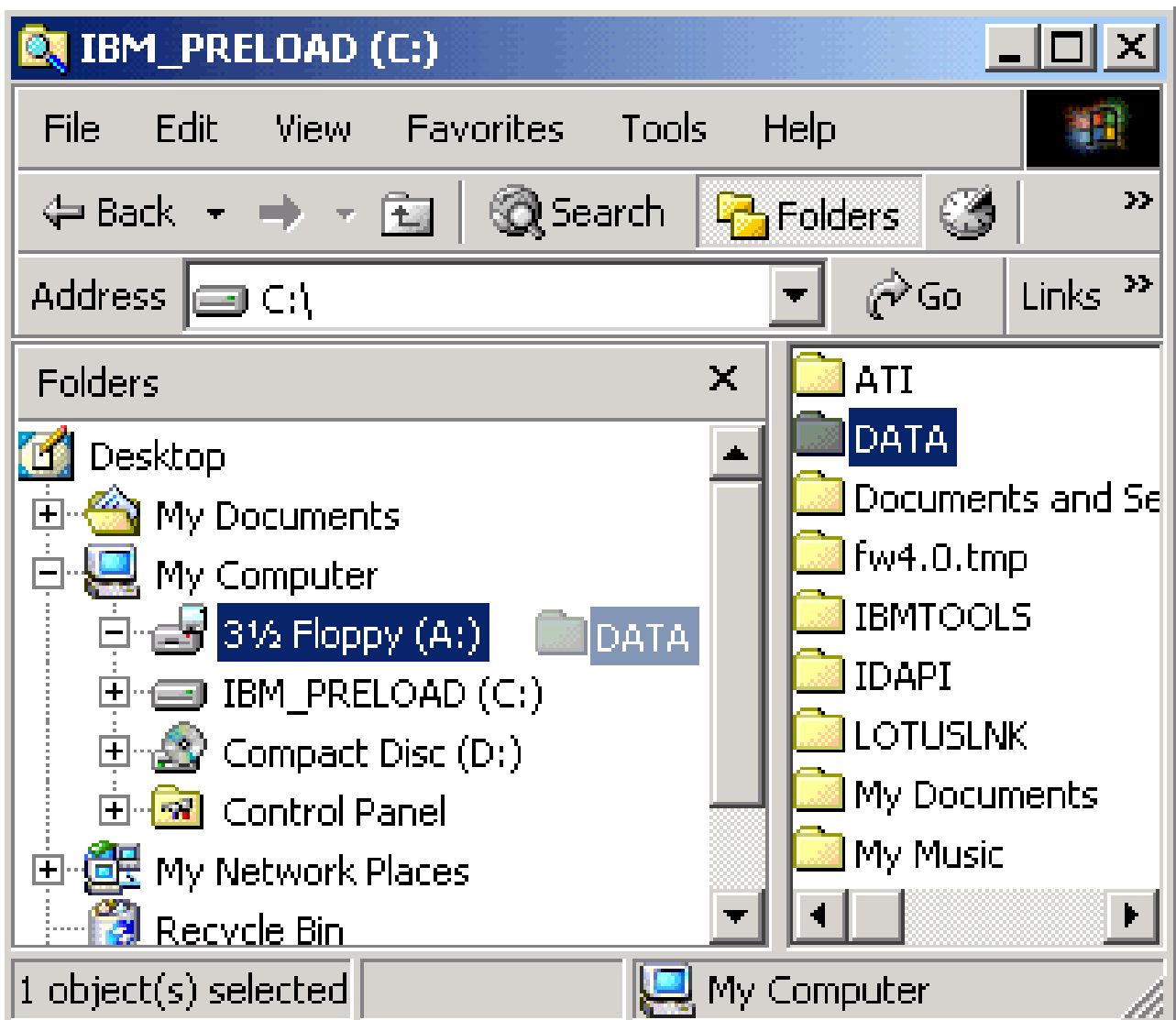
8. Drag and Drop

Drag is when the mouse button is held down while the mouse is moved. It is mostly used to move files around in Windows Explorer or in some other applications. Drop is the release of the button, after finishing dragging the mouse. Here is an example of dragging and dropping a file from the hard drive to the floppy drive in Windows Explorer:

- ♦ Select the file, hold down the left mouse button and drag it in the direction of the floppy drive on the left side.



- ◆ After the floppy drive highlights you can release the mouse button; the file will be dropped and copied there.



9. File Extensions

A file extension is the three letters of a file name after the dot. It is created by the application when you save a file. It is the way operation systems identify which application to use to open a file. Windows often doesn't list them when you are searching for a file, but they are there. Examples are: .txt, .doc, .exe, .html, .jpg, .gif, .wav.

However, sometimes you will want to save a file with different extensions. In order to do that, select Save As in the File menu. The Save As box will open. Choose from the drop-down menu next to Save As Type the type you want to save. The extension of the type will be created for you by the application.

Here is an example from Microsoft Word:



Some file extensions:

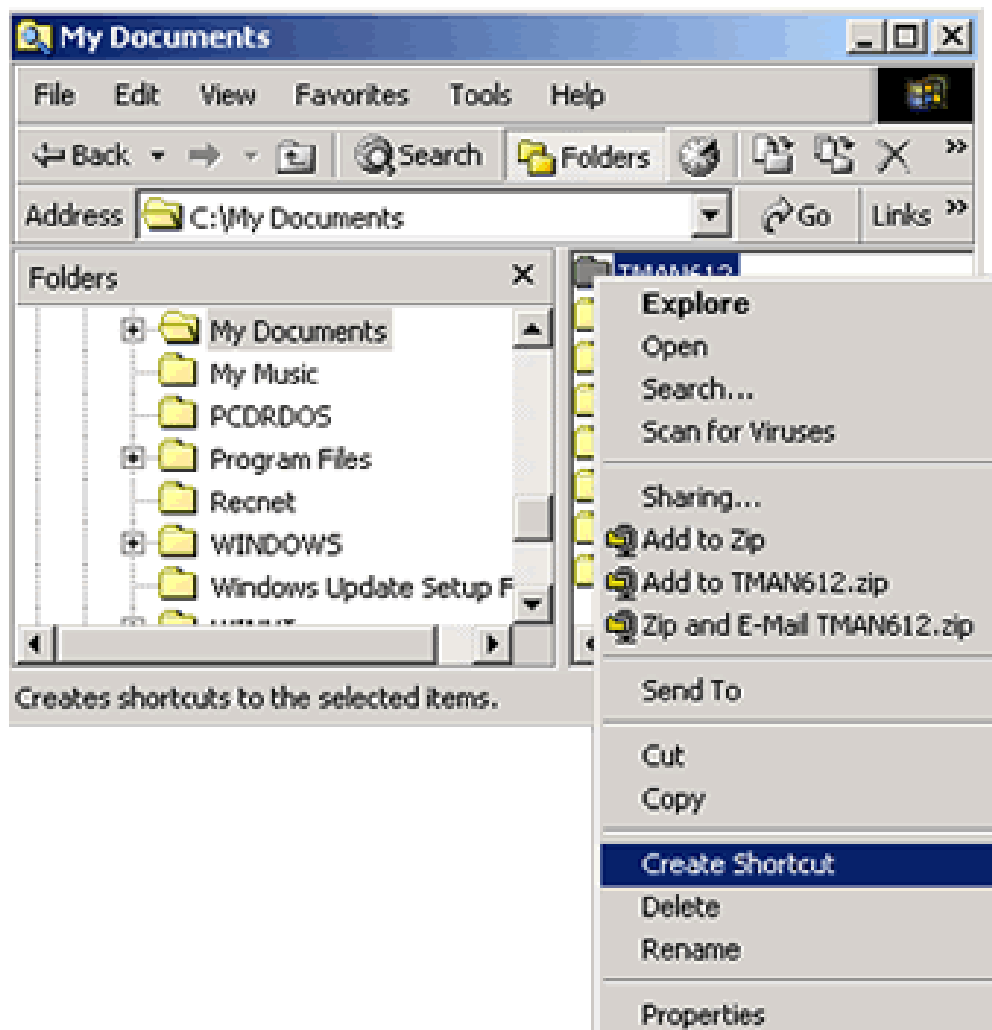
- ◆ Graphic files: **.bmp, .gif, .tif, .jpg**
- ◆ Word document: **.doc**
- ◆ Program file: **.exe**
- ◆ WebPages files: **.htm, .html**
- ◆ Unformatted text: **.txt**
- ◆ Excel spreadsheet file: **.xls**
- ◆ Compressed file: **.zip**
- ◆ Rich Formatted Text: **.rtf**
- ◆ Sound file: **.wav**
- ◆ Acrobat Reader file: **.pdf**

10. Shortcut

A shortcut creates a button or icon which typically stays on the desktop and when clicked, quickly allows you to start a program or open a file without having to go to its permanent location on your computer.



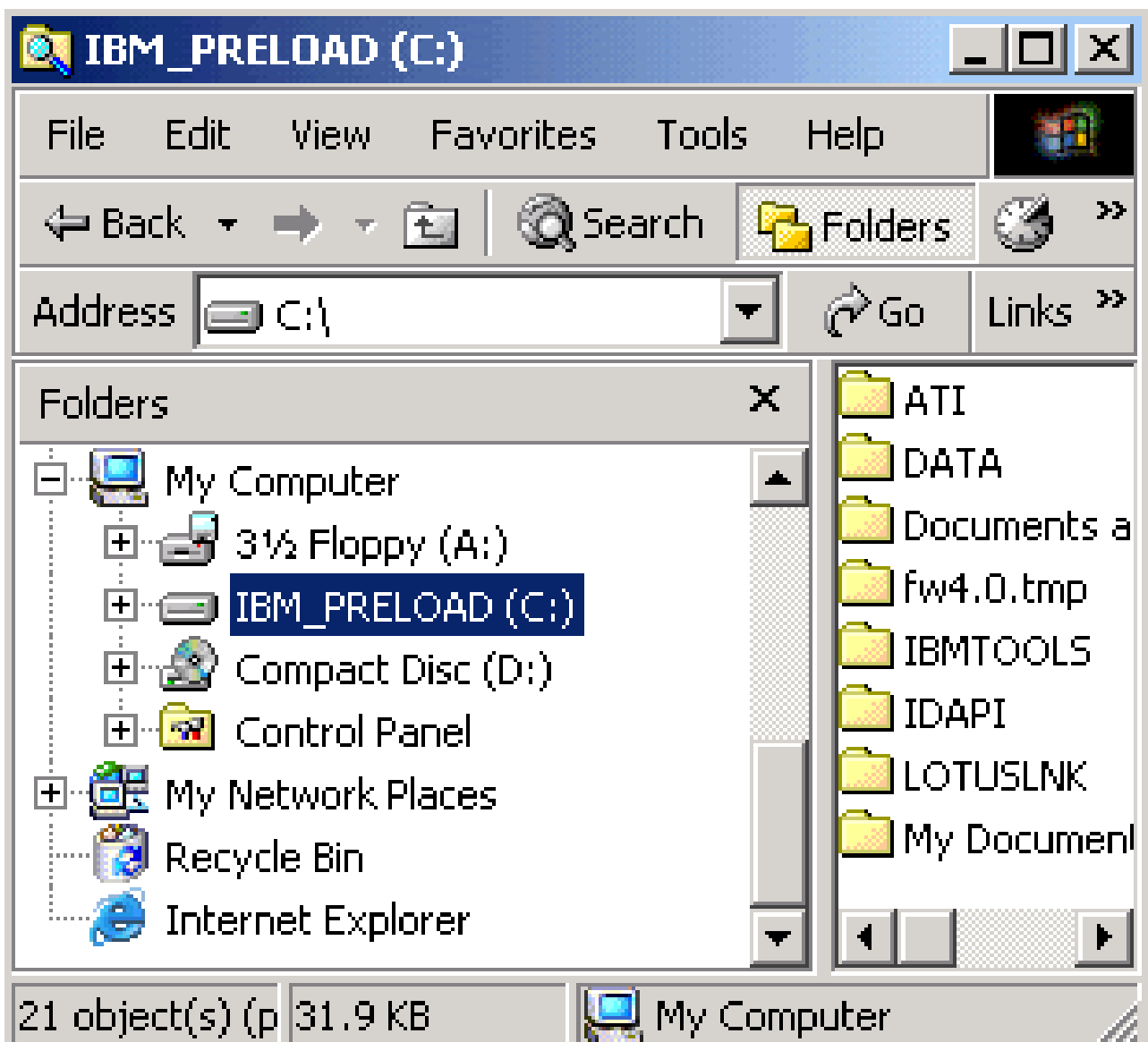
To create a shortcut, open the Windows Explorer or go to the Start menu, select the file or program you want to create a shortcut for, place the cursor on the icon for the file, then right-click and select Create Shortcut from the menu that pops up. The shortcut file will be created. After that, you drag the shortcut file to the desktop and whenever you click on it, it will open the application or document related to the shortcut.



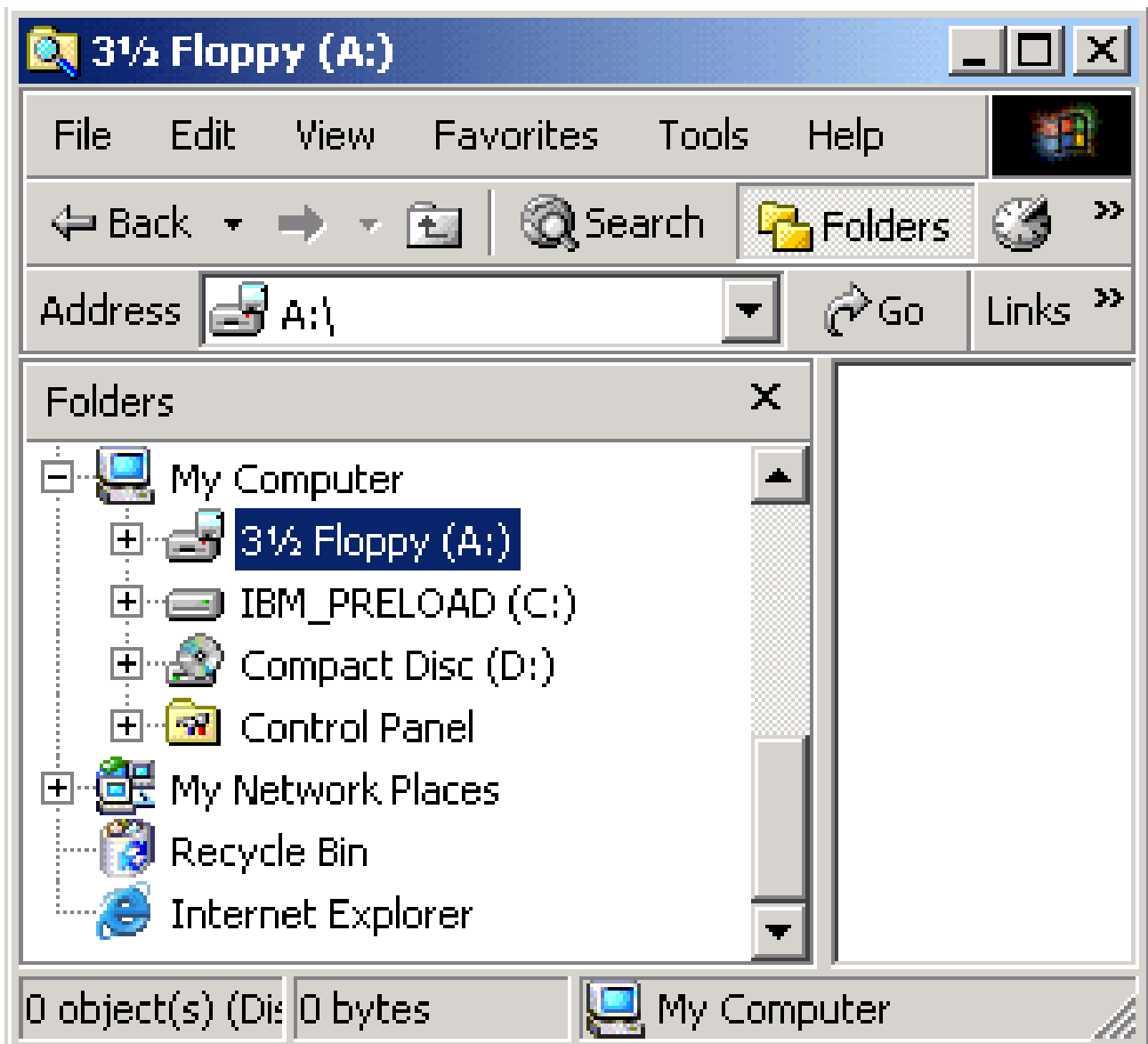
11. Hard Drive and Floppy Drive

In your computer there are different drives: the hard drive, the floppy drive and the compact disk drive.

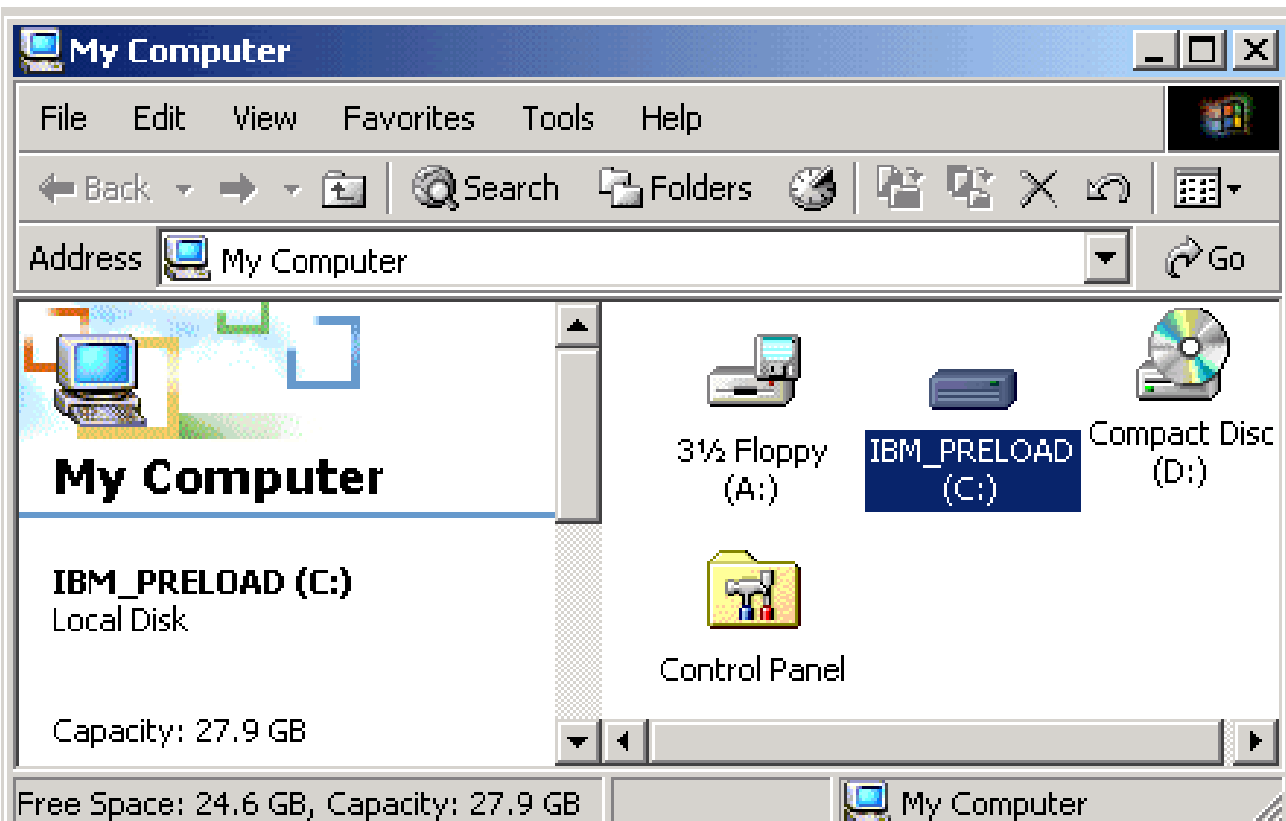
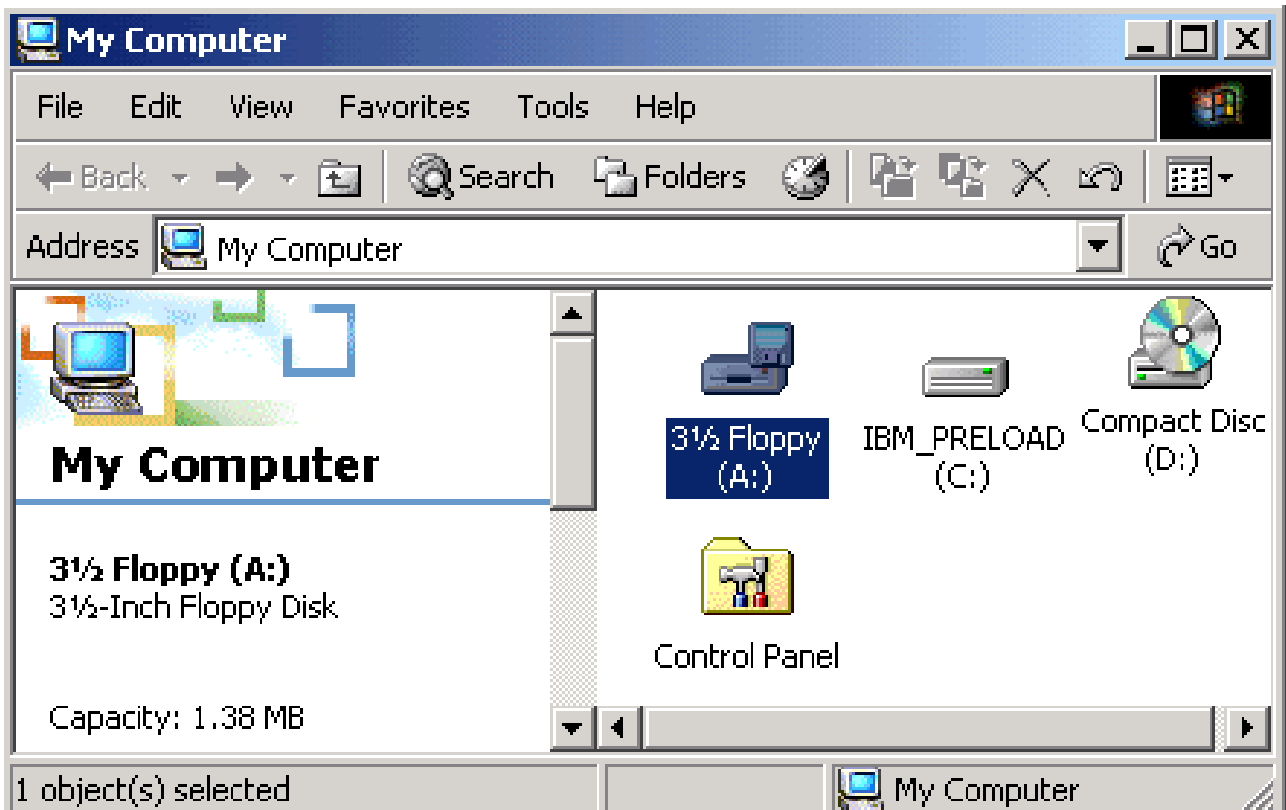
- ♦ The hard drive, also called the C: drive, is used for storing all your applications and files.



- ♦ The floppy drive, or A: drive, is used when you insert a floppy disk to store files that you want to transport or create a back up.



or in My Computer, select the Hard or Floppy Drive.

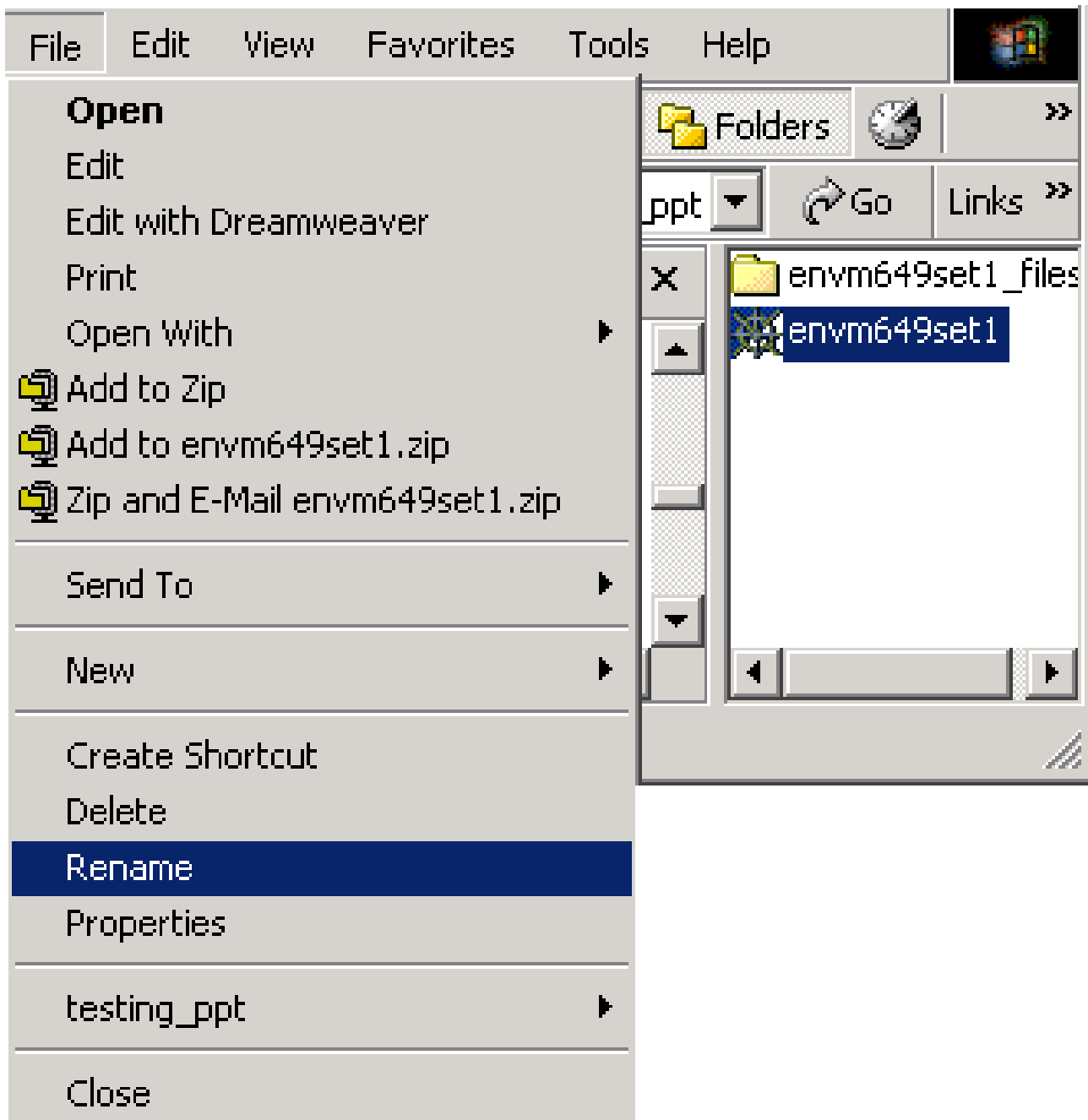


12. Rename/Delete a Folder or File

Using Windows Explorer you can rename or delete a file or folder.

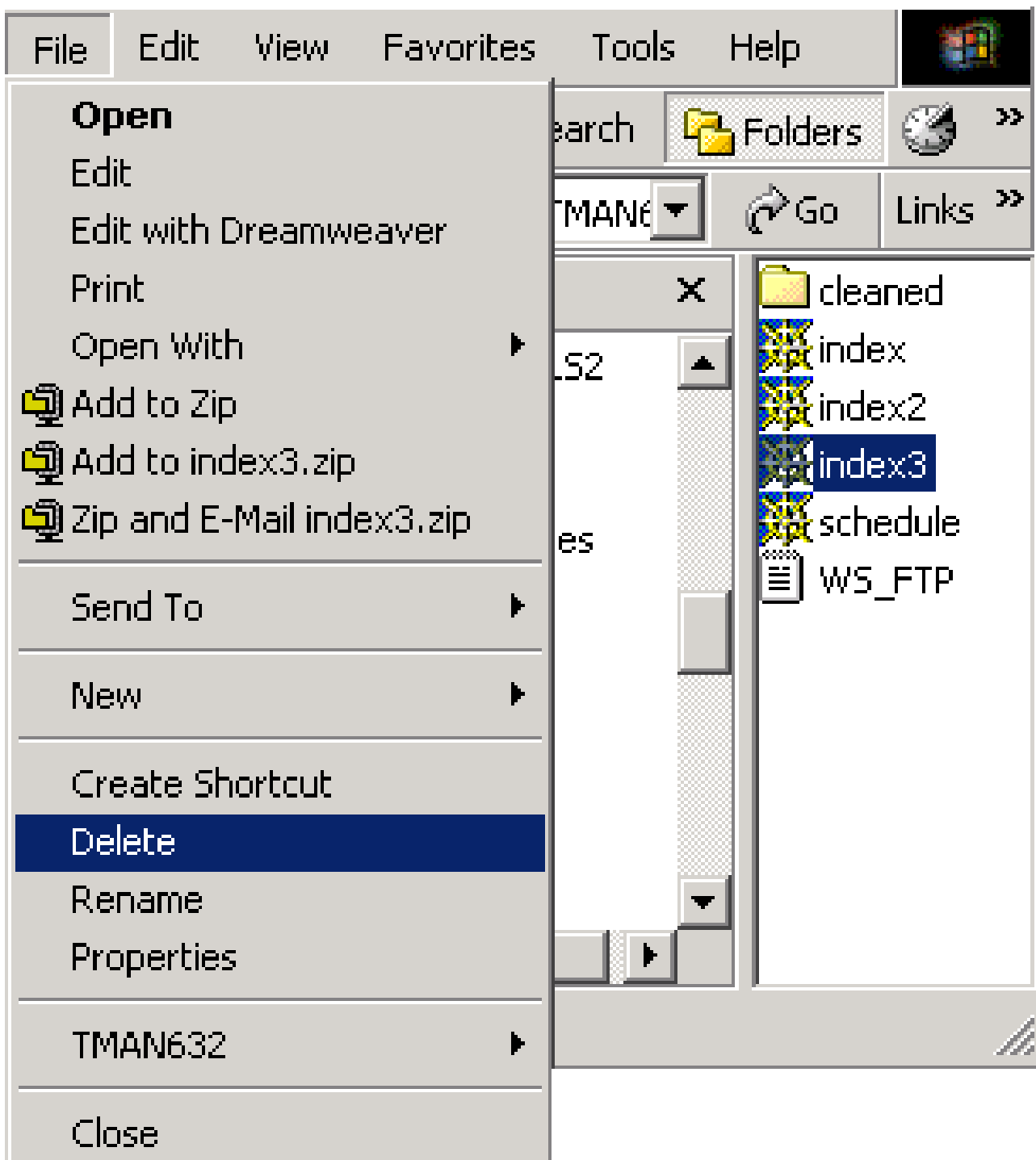
To Rename a file or folder:

- ♦ Navigate and select the file or folder you want to rename from the proper drive (c:, a:).
- ♦ Go to the File menu and select Rename.
- ♦ The file name will be highlighted and blinking in the right corner waiting for you to type the new file name.



To Delete a file or folder:

- ♦ Navigate and select the file or folder you want to delete.
- ♦ Go to the File menu and select Delete.



Another way of renaming or deleting a file or folder is using the right-click button of your mouse. After highlighting the file or folder, you right-click and a menu will appear where you can select Rename or Delete.

13. Recycle Bin

When you delete a file from your computer's hard drive, it is removed to the folder called Recycle Bin. Files deleted from your hard drive are still available for restoring if needed, unless you empty your Recycle Bin.

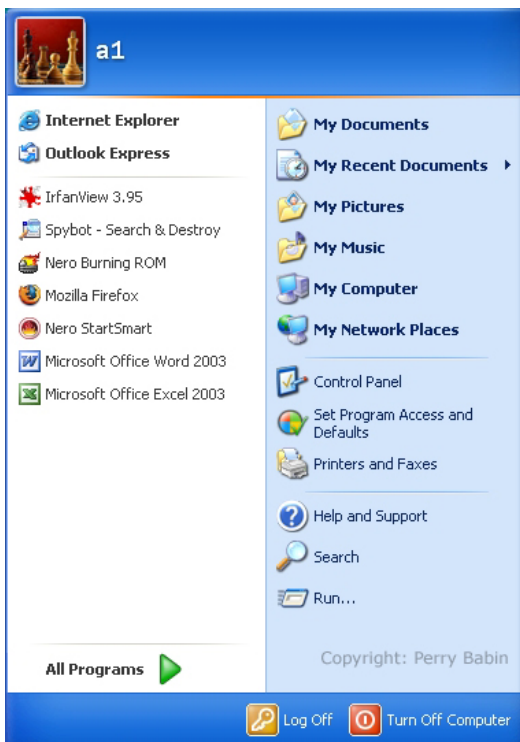
- ♦ To restore the files, open the Recycle Bin folder by double-clicking the icon on your desktop, select the file and then select Restore.
- ♦ To clear the Recycle Bin, select Empty Recycle Bin and the files will be permanently removed from your computer.



Note that the recycle bin does not apply to the contents of a floppy disk. If you delete a file from a floppy disk, it is permanently deleted, and you cannot restore it.

14. Customized basic default settings

The list at the lower left of the dialog box contains the most frequently used programs. They are arranged in the order of most used to least used. The total number of items to be displayed in the list can be set by RIGHT-clicking on a clear area of the start menu and selecting PROPERTIES. From there, select the 'start menu' tab. You will see a dialog box similar to the following:

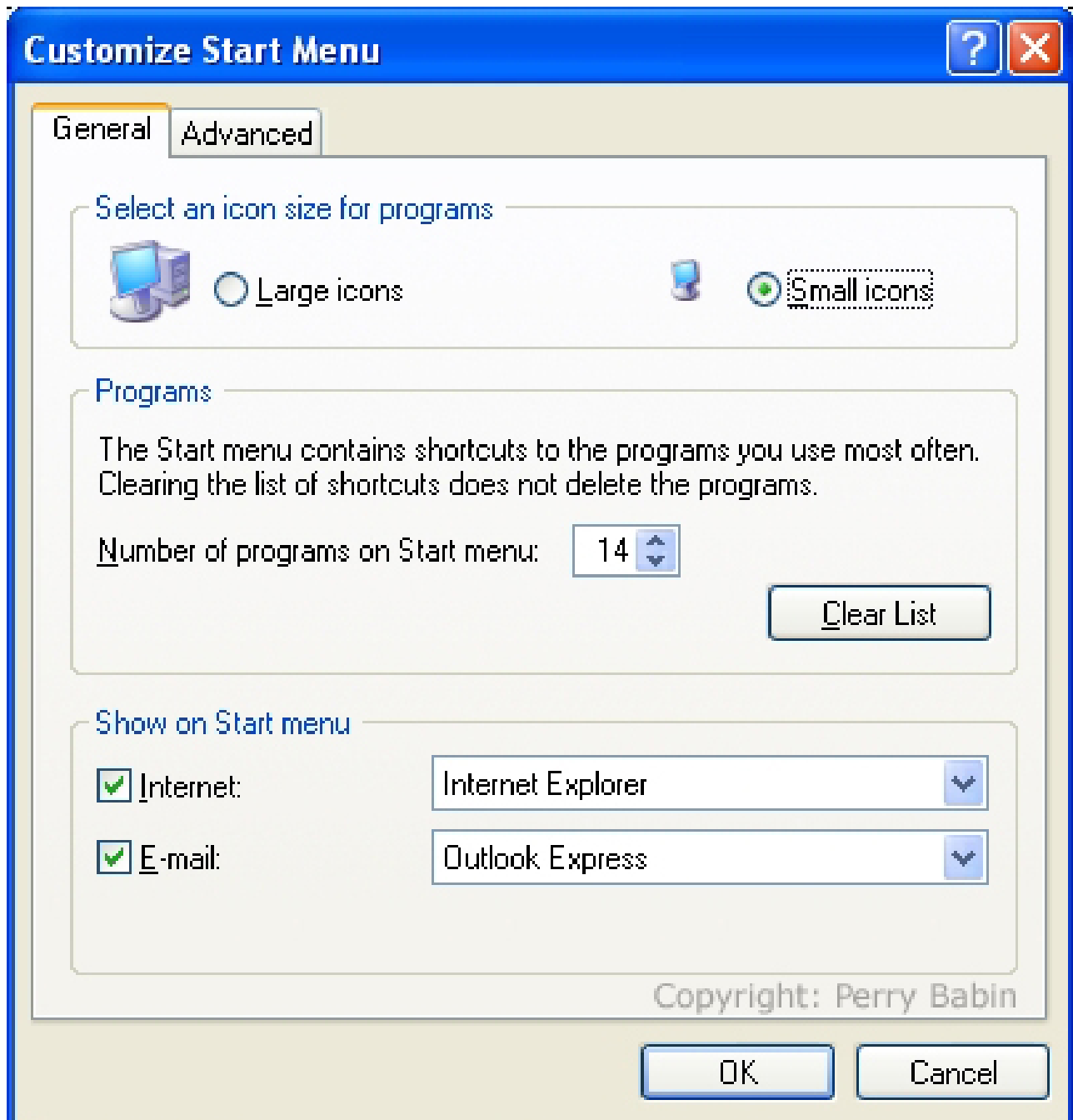


Here, click the 'customize' button.

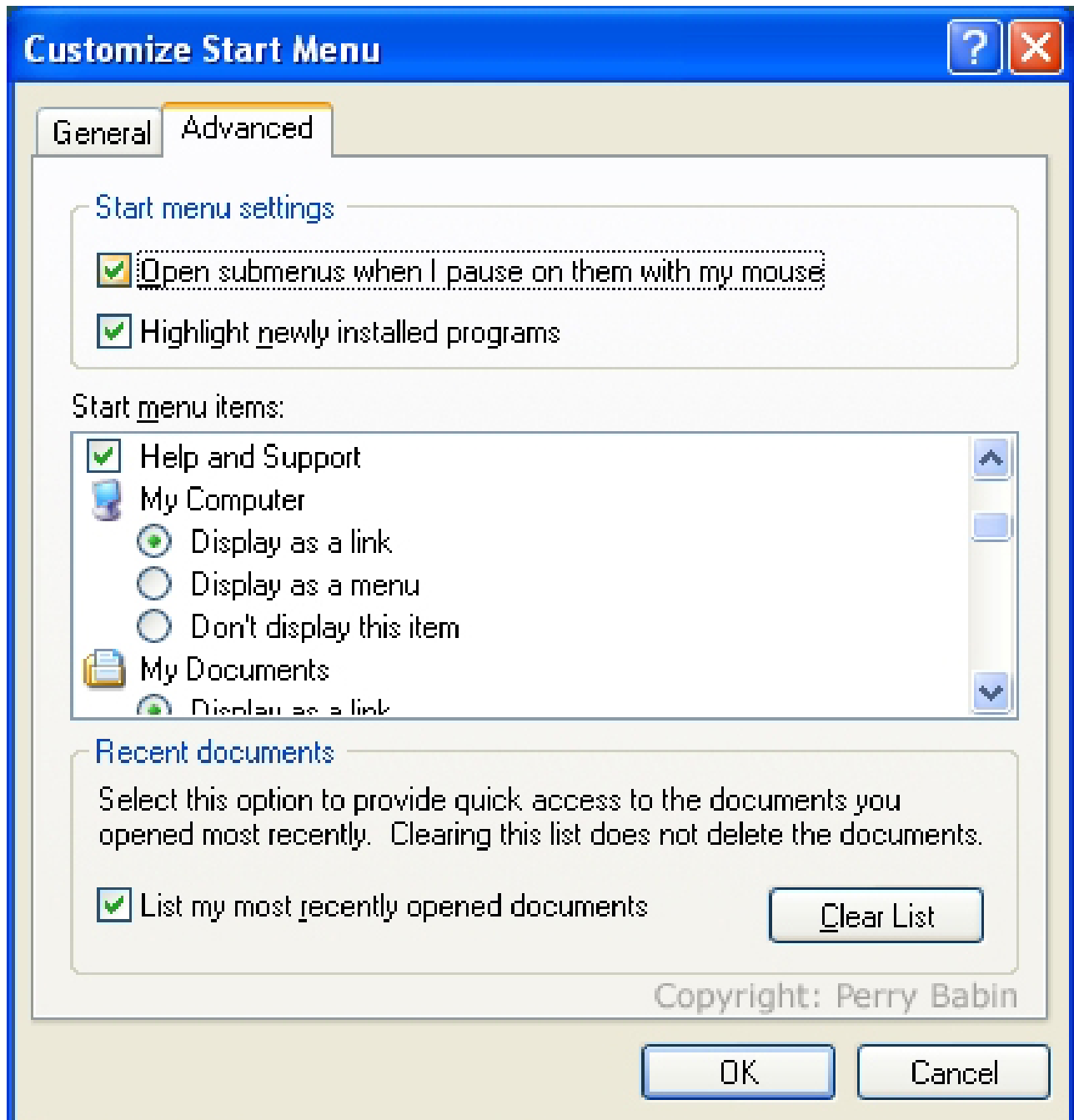


On this tab, you can set the maximum number (limited by the area available in the menu). You can set the size of the icons. I suggest that they should be set to small unless you have vision problems. You can also tell it whether you want links to the your internet browser and email client. If you use the pull-down menu for the browser/email programs, you can select the link that will be displayed. When you finish making your selections, click the OK buttons to save your settings and to close the dialog boxes.

The 'clear list' button here will clear the list of the most frequently used programs. If you will regularly be clearing the frequently used programs menu but you'd like to keep a couple of programs to permanently reside on the start menu, you can right-click the program and select 'pin to start menu'.



On the 'advanced' tab, you have more selections. If you're unsure about any of the choices, you should leave them as they are. If, at any time, you change something and you can not remember how to set it back as it was, simply click 'cancel' on all of the open dialog boxes and no setting will be changed. The 'clear list' button here will clear the list of the most frequently opened documents.



15. Basic software programs

15.1 Microsoft Excel

Microsoft Excel is a spreadsheet programs that is primarily used for calculation and analysis. It is a very powerful tool. Below is s screenshot of a blank page. As you can see, there are rows (left/right) and columns (up/down) of cells. These cells will generally contain either numbers or formulas. The formulas can be as simple as adding the values of two cells.

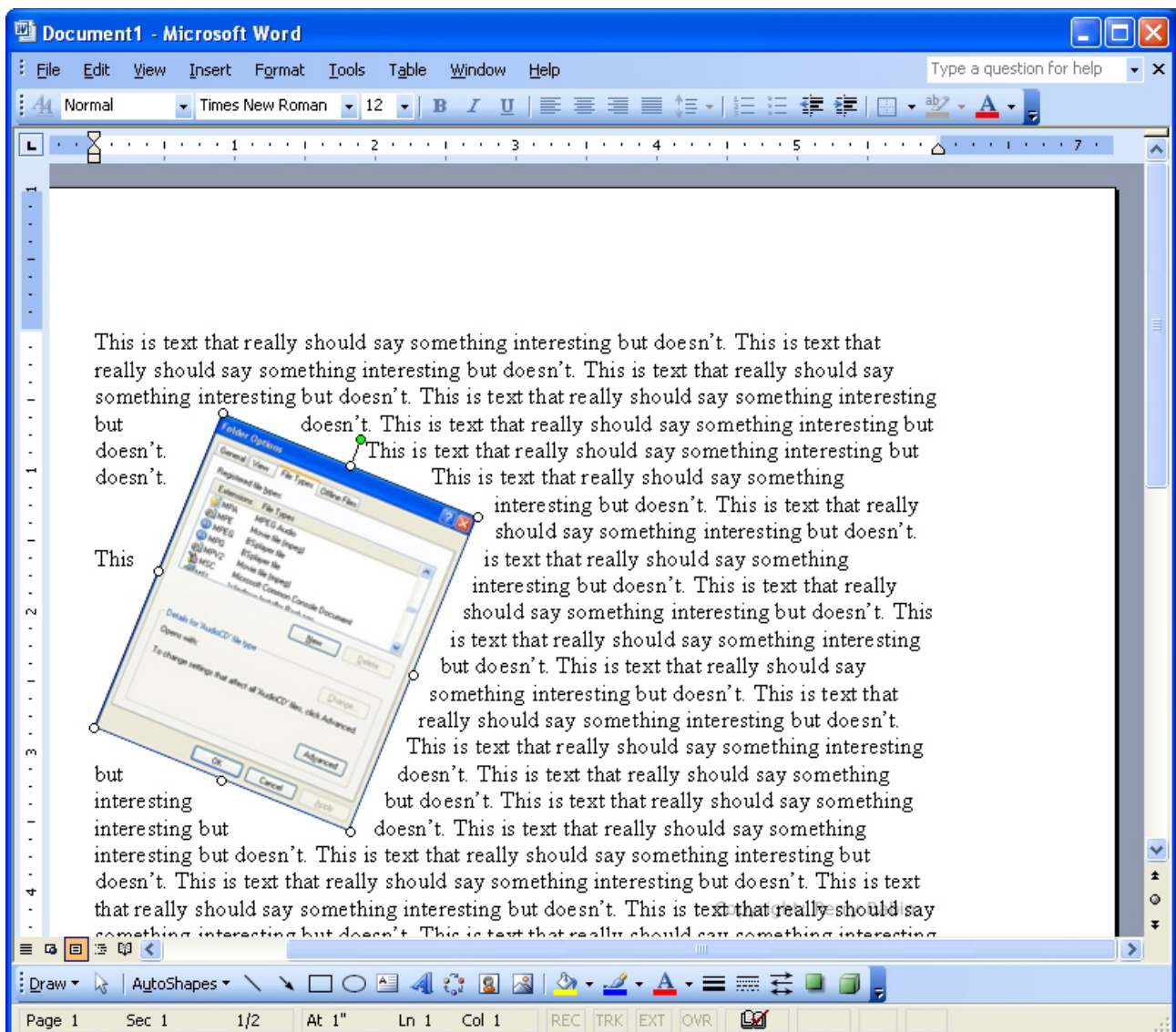
The screenshot shows a Microsoft Excel window titled "Microsoft Excel - formatting.xls". The spreadsheet contains the following data:

Quantity	Part #	Description	Back Ordered	Shipped	Each	Discount %	Before Discount	Extended
25	MB-123	Motherboard	2	23	\$38.95	10%	\$895.85	\$806.27
12	MB-246	Motherboard	0	12	\$38.95	5%	\$467.40	\$444.03
5	MB-369	Motherboard	0	5	\$38.95	0%	\$194.75	\$194.75
							Subtotal	\$1,445.05
							Tax 8%	\$115.60
							Total	\$1,560.65

Copyright: Perry Babin

Microsoft Word is a word processor/text editor. Other similar programs are Windows' Notepad and Windows' Word Pad. Word is by far the most powerful of the three. It has many more features than the casual user will need. I will only cover the basics here. Below, we see a blank page.

Below the main toolbar (with File, Edit...), we see the formatting toolbar. On the left, you can see the 'styles and formatting' drop-down menu. This contains some commonly used styles. Here you can quickly select a style of text that you use often. The styles and formatting are customizable and you can add to the list.



16. The online help on Windows

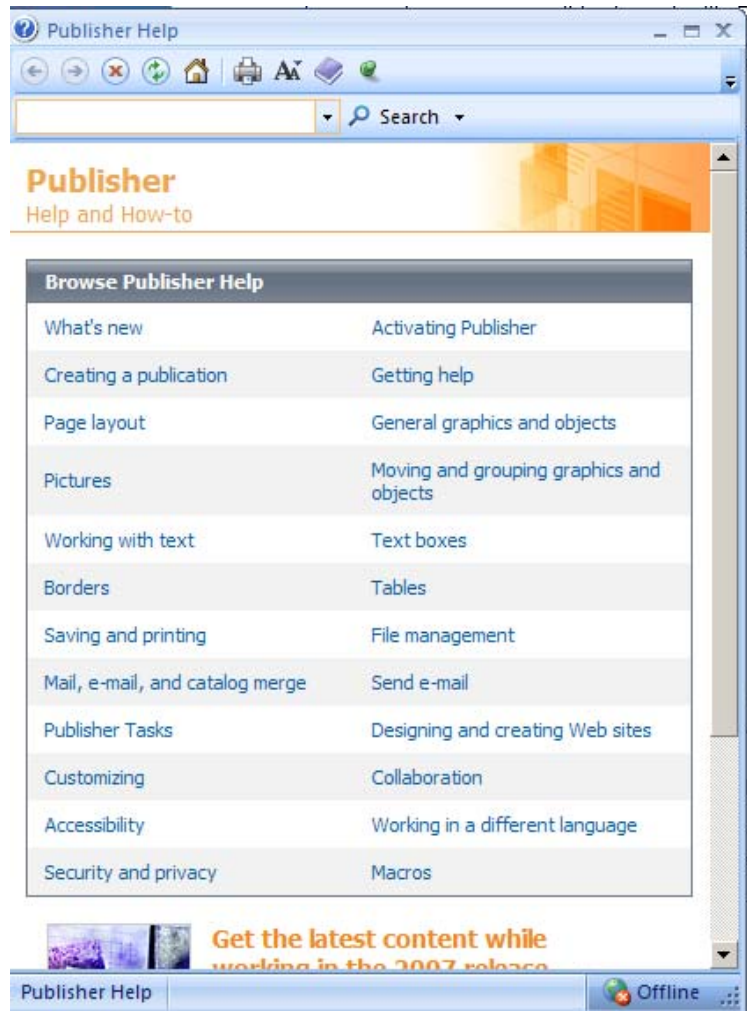
(Online help means help that's built into the computer. Offline help would be hiring a consultant, reading a manual, asking a friend.) In Windows there's a Help menu at the end of every windows application menu. Pull this down for online (i.e.: in the computer) help. The Windows system itself offers Help through the Start menu

Getting more help

- ◆ All programs offer online help
- ◆ Most online help contains
- ◆ A Find feature
- ◆ An Index feature
- ◆ More narrative style of help

With most help may be able to:

- ◆ Print out help topics
- ◆ Annotate help
- ◆ Mark favorites or bookmark help topics you need frequently



[illegible]

STUDY UNIT 6

Basic maintenance on a PC

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Identify the top ten computer maintenance solutions
- ♦ Perform a basic trouble shooting exercise.
- ♦ Demonstrate decision-making skills by referring a problem to the correct person within your company.
- ♦ Identify the correct method to remove viruses on the PC.
- ♦ Distinguish between abusing a PC and misusing a PC.

1. The Top 10 of Basic Computer Maintenance

- ♦ **Backup, backup backup!** Backing up your data just means saving your files to a separate location (like a cd, floppy or zip disk) so in event of disaster you can easily copy your files back onto your computer. How often really depends on you. Just ask yourself, "If I only back up once every 3 months, would anything I saved on my hard drive since the last time I saved it on a backup be all right if I could NEVER get it back?"
- ♦ **Invest in an UPS (uninterruptible power source).** A UPS is a wonderful device that works as a surge protector and a battery backup all in one! Not only does it keep your computer safe from electrical spikes that can be traveling through the lines, it also gives you time to save your work, and properly power down your computer in event of a power failure. You don't lose your work, and your computer gets to be shut down properly.
- ♦ **Never turn the computer off with the power switch.** The one exception to this rule is when your computer locks up and your hard drive is not running (the hard drive light is not blinking). In this situation, you can turn the power off without harmful effects to the hard drive. As cutting the power can also result in lost data or Windows files, you should only do this when you have to.

Doing this can help prevent permanent hard drive damage caused by the hard drive heads contacting the surface of the drive disc. This can prevent a host of Windows problems.

Whenever possible, recover from crashes by pressing the Ctrl + Alt + Delete keys at the same time and going into your Task Manager. You can often find programs that are not responding in here, and chose to close them from this utility. Press Clt + Alt + Delete again to reboot your computer from this screen.

- ♦ **Scan disk and defrag once a month.** These utilities help to keep your hard drive in optimal shape and error free.
- ♦ **Do not unplug peripheral devices while the power is turned on.** Unplugging devices such as your mouse, keyboard, printer etc with the power on can short out the connector socket or even the motherboard itself. The only time you would unplug something while the power is on, is if the device is hot swappable/pluggable. If you are unsure if something is hot swappable or not, power down first to play it safe.
- ♦ **Remove programs from your startup.** Often when you install a program, it will ask you whether it should load up as soon as Windows loads (or just decide to do so on its own). The more programs you have trying to start up at the same time, the slower your computer will start. Also, all those programs running in the background while you try to work will slow your computer's response time. In windows 98, ME, and XP, you can do this by going to your Start -> Run -> in the command box type in 'msconfig' and a small window will appear. Go to the tab marked "Startup" and you can chose what items to remove from there. In Windows 2000, you can do this by going to your Control Panel -> Administrative Services -> Services.

- ♦ **Run virus checks and adware/spyware scans regularly (at least once a week).** There are so many types and variations of viruses, spyware and adware floating around that at times it may seem daunting to even have internet access on your computer. However, regular scans with your virus scanner, and adware scanners, can ensure your computer remains healthy.
- ♦ **Keep disks handy.** If the worst thing happens, and you need to format your system for one reason or another, then it is a good idea to have all your driver disks, operating system disks, backup disks and so forth in a readily available place. Running around in a panic at the 11th hour because you cannot remember where the drivers are for your network card, is not a good place to be.
- ♦ **Keep your software updated.** Software developers are always writing and releasing patches that update the software you have already bought from them. A lot of these patches fix problems you may be experiencing yourself with a particular program. Other patches plug security leaks. Still other patches update reference files for your virus/spy/adware scanners to make sure you are scanning for as many variants as possible. So make sure you regularly update/patch your operating system, virus scanners, and various other applications!
- ♦ **Develop a schedule of maintenance practices.** All this advice is meaningless if you keep forgetting to do it, or only remember after something has gone wrong. Write yourself up a maintenance schedule, with a list of what tasks should be performed daily, weekly, and monthly. If you develop a routine and stick to it, then you should avoid the larger problems by always being on top of the smaller ones!

2. Basic Troubleshooting

Troubleshooting computers is like solving the problems you solve in everyday life. Once you understand the basic structure, you can easily determine where the problem lies.

2.1 What if there appears to be no power?

If your computer fails to power up (you can hear no fans running and no lights are lit when you press the power button), check the following:

- ♦ Make sure that you're pressing the power button on the front of the computer and not the reset button.
- ♦ Make sure that the power cord is securely plugged into both the power outlet and the computer's power supply.
- ♦ Make sure that the main power switch on the back of the computer's power supply is on.
- ♦ Make sure that the outlet you have the computer plugged into has power. Plug a lamp into the socket that the computer is plugged into. If the lamp lights, the outlet has power. If the lamp does not light and it's plugged into an outlet strip, plug the lamp into the wall socket. If the lamp lights when plugged directly into the wall socket but not when plugged into the outlet strip, be sure that the strip's power switch is on. If it has a circuit breaker, be sure that the breaker has not tripped.

If the lights light up when you press the power button but you get no display, check the following:

- ♦ Be sure that the monitor is on.
- ♦ Confirm that you have power to the monitor.
- ♦ If you have an LCD monitor, there may be a small power supply (typically a small rectangular black box) between the wall plug and the plug that connects to the monitor's power supply input. Be sure that the plugs on the power supply and monitor are securely seated.
- ♦ Make sure that the monitor is plugged in securely to the video connector on the back of the computer. This is covered on page 3 of the tutorial.
- ♦ If the monitor still has no display, see if the on-screen display for the monitor's controls can be displayed. You may need to find the owner's manual for your monitor to learn how to access the on-screen display.



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If the computer and the monitor have power but you still cannot get it to operate, you may want to have it checked by a computer tech.
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3. Remove Spyware and Viruses

Use antivirus and spyware detection software to remove malware from your computer. If you suspect that your computer has been infected by adware, spyware, viruses or a Trojan you should run your antivirus and spyware removal programs.

Anti-virus programs such as Symantec, McAfee, AVG and Avast can often remove the virus by allowing you to quarantine and delete infected files. And spyware removal programs such as Ad-Aware and Spybot Search and Destroy can remove adware and spyware by allowing you to quarantine and delete unwanted tracking programs. But, sometimes viruses, especially Trojans, are harder to remove and your anti-virus and spyware removal programs will not be up to the task.

3.1 Determine the Virus' Identity

If your computer is infected by a Trojan or other difficult to remove virus your anti-virus or malware removal program should be able to detect the presence of the virus even if it can't remove it. When these programs find an infected file they display the file name along with the virus name. You may notice that even after a second run of the program it is picking up more files infected by a certain virus. Or, your program may display a special alert window that a specific Trojan has been found on your computer. In either case you will want to write down the name of the virus or Trojan found on your computer. Having the name of the virus is the first step in removing it.

3.2 Using Virus Removal Tools

If possible, use a different computer for these next steps. First, go to Symantec's website and search their virus removal tool database for the virus found on your computer. Symantec keeps specialized removal tools for new and common viruses and Trojans available free for download. If you find a removal tool for your virus in their database download it to install and run on the infected computer. Any special instructions will be provided at the download site. These instructions may have you disconnect your infected computer from the Internet and boot it up in safe mode. Running the removal tool will remove the specified virus or Trojan.

3.3 Virus Removal Help

If you cannot locate a virus removal tool for your problem virus or Trojan or the tool is outdated and does not work then you may want to try searching an online website or forum for virus removal help. PC Hell and Bull Guard are great help sites for virus removal. Search these sites for references to the virus or Trojan infecting your computer. Step by step instructions may have already been posted by others. Or, post your own question asking for help in removing the virus. Another option is to use a search engine such as Google, Dog pile or MSN search to find references to the virus or Trojan online. This may provide you with alternate removal tools, removal documentation or better information about the virus or Trojan.

3.4 Removing the Virus

Once you have a removal tool or written steps for removing the virus or Trojan, go back to the infected machine. Most likely you will need to disconnect the computer from the Internet before trying to remove the virus. Follow the removal instructions you have found about the specific virus to the letter. If you have any questions about these instructions go back to the website and ask before attempting to remove the virus. Some instructions may call for you to delete registry values. If you are not comfortable doing this advanced procedure you may want live online or phone support. In some cases you can take your computer to locations such as Best Buy and Circuit City for virus removal help.

4. Abuse to and Misuse of the Computer

Abuse: This refers to physical abuse and neglect of computer equipment. Examples of this are banging the keyboard, kicking the CPU, hitting the screen, etc.

Misuse: This refers to the computer being used for anything other than business and used to do personal things, for example typing of private documents, playing games, e-mailing friends, etc.

5. Problems and Solutions

	Solution
Hardware Failure	Contact the service agent for your particular equipment. Make sure you obtain this information and keep it handy.
Software freeze	Restart the computer. You may lose some work, but there is no other way to solve the problem. Make sure that you have set up the auto-recovery options on the programs you are working on.
Back-up Failure	Contact the company IT support officer for your office/ / region.
Alternative Ideas	There may be circumstances where the computer is not working due to a power supply disruption or the computer is malfunctioning. You will then need to make alternative arrangements or have a manual system in place for certain tasks such as, sending letters, registers etc.

[illegible]

WORDS/ACRONYMS	EXPLANATION OF ACRONYMS
ASSESSMENT	“A structured process of gathering evidence and making judgments about an individual’s performance in relation to registered national standards and qualifications”
ASSESSMENT CRITERIA	It describe the evidence that is needed to show that the learner has demonstrated the outcome correctly. A list of evidence to be supplied by the candidate as proof of his / her ability to integrate all of the competencies listed earlier in the standard.
COMPETENCE	Competence is the ability to do certain things in a range of different environments with knowledge and understanding about what is being done and the effect this has on others. It includes <u>Knowledge, skills, attitudes, values and ethics.</u>
CREDIT	Each unit standard is worth a number of credits which gives an indication of the size and scope of the unit standard, the standard is worth. (1 credit = 10 notional hours of learning)
CRITICAL CROSSFIELD OUTCOMES	critical cross field outcomes is to direct educational activities towards development of the learners within a social and economic environment.
EMBEDDED KNOWLEDGE	Body of theory essential for understanding and applying the standard. Principles, methods and techniques that would underpin good practice of this unit standard.
EVIDENCE	Evidence is a set of information or observations that provides proof of the learner’s competence.
FORMATIVE ASSESSMENT	Aims to assist the learner to identify any knowledge/skills gaps when demonstrating his/her current competence. Diagnostic assessment supports learning and progress of the learner.
INTEGRATED ASSESSMENT	Integrated assessment refers to: • assessing a number of outcomes together • assessing a number of assessment criteria together • assessing a number of unit standards together • using a combination of assessment methods and instruments for an outcome/s • collecting naturally occurring evidence in the workplace • Acquiring evidence from other sources such as supervisor’s reports, testimonials, portfolio of work previously done, logbooks, journals etc.
LEARNING ASSUMED TO BE IN PLACE	Building blocks (knowledge, skill and understanding) assumed to be in place
LEARNING PROGRAM	A “learning program is a set of learning activities in which the learner will be involved in working towards the achievement of one or more outcomes.
LEVEL	Each unit standard is registered on the NQF at a particular level based on the complexity of the outcomes and stringency of the standards.
NQF	National Qualifications Framework
RANGE STATEMENTS	The range statements indicate detailed requirements of the assessment criteria.

RE-ASSESSMENT	• The same method and instrument may be used, but, the task and materials should be different • The task and materials should, however, be of the same complexity and level as the previous ones. • In case the methods and instruments are changed it must be ensured that they are appropriate for the outcomes specified.
RPL ASSESSMENT	RPL is a way of recognizing what individuals already know and can do. The basic premise underlying it is that people, especially mature adults, learn many things inside and outside the formal structures of education and training and that irrespective of where, how and when learning was acquired, subject to assessment, it can be worthy of recognition and credit.
RPL	Recognition of prior learning
SASSETA	Safety & security sector education and training authority.
SAQA	SOUTH AFRICAN QUALIFICATION AUTHORITY
SPECIFIC OUTCOMES	An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention. Key performance areas in the standard are given, the focus is <i>not</i> only on current competence, but also on future competence in this standard.
SUMMATIVE ASSESSMENT	Assessment used to record the level of competence achieved by a learner over a period of time, e.g. at the end of a program, term or year.
QUALIFICATIONS	Represent a planned combination of learning outcomes which has a defined purpose and intended to provide learners with applied competence and a basis for further learning
UNIT STANDARDS	the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF). “A statement of the outcomes that the learner must achieve to earn credit. Standards are nationally agreed and internationally comparable.”

- ♦ <http://cuip.uchicago.edu/wit/2000/curriculum/homeroommodules/compBasics/CBMouse.htm>
- ♦ <http://www.microsoft.com/windowsxp/using/setup/maintain/filemgmt.mspx>
- ♦ <http://computer.howstuffworks.com/operating-system2.htm>

Operate a computer workstation in a business environment

SAQA US ID	UNIT STANDARD TITLE		
114979	Operate a computer workstation in a business environment		
ORIGINATOR		REGISTERING PROVIDER	
SGB Insurance and Investment			
FIELD		SUBFIELD	
Field 03 - Business, Commerce and Management Studies		Finance, Economics and Accounting	
ABET BAND	UNIT STANDARD TYPE	NQF LEVEL	CREDITS
Undefined	Regular	Level 3	2
REGISTRATION STATUS	REGISTRATION START DATE	REGISTRATION END DATE	SAQA DECISION NUMBER
Reregistered	2007-06-26	2010-06-26	SAQA 0160/05
LAST DATE FOR ENROLMENT		LAST DATE FOR ACHIEVEMENT	
2011-06-26		2014-06-26	

PURPOSE OF THE UNIT STANDARD

This Unit Standard is intended for learners who use a computer in a business environment. It is an introduction to basic computer literacy.

The qualifying learner is capable of:

- Explaining basic IT concepts.
- Identifying and operating the hardware in a specific workstation.
- Using a computer operating system.
- Demonstrating basic maintenance procedures.

LEARNING ASSUMED TO BE IN PLACE AND RECOGNITION OF PRIOR LEARNING

Learners should:

- Hold a GETC or equivalent qualification
- Be competent in Communication and Mathematical Literacy at NQF Level 2

UNIT STANDARD RANGE

The typical scope of this Unit Standard is:

- Navigating the operating system includes, but is not limited to, menus, toolbars, buttons, short cut keys and drag and drop.
- Basic maintenance includes archiving, deleting, defragmenting and using the maintenance facility of a workstation.

UNIT STANDARD OUTCOME HEADER

N/A

Specific Outcomes and Assessment Criteria:

SPECIFIC OUTCOME 1

Explain basic IT concepts.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

The concepts hardware and software are explained and an indication is given of when and how they work together.

ASSESSMENT CRITERION 2

The concept of a computer virus is explained with examples.

ASSESSMENT CRITERION 3

Policies relating to IT in a specific organisation are interpreted and an indication is given of the consequences of non-compliance.

ASSESSMENT CRITERION 4

The difference between a stand alone and network system is explained and an indication is given of the implications of each type of system for the user.

ASSESSMENT CRITERION 5

The security access in a particular organisation is explained with references to policies and standards.

ASSESSMENT CRITERION 6

The importance of backing up is explained with reference to the standards of a specific organisation.

SPECIFIC OUTCOME 2

Identify and operate the hardware in a specific workstation.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

The hardware connected to a specific workstation is identified and an indication is given of where the peripherals are located in the building.

ASSESSMENT CRITERION 2

The computer is switched on and the keyboard or mouse is used to log on.

ASSESSMENT CRITERION 3

The use of the keyboard and mouse is demonstrated for a specific purpose.

SPECIFIC OUTCOME 3

Use a computer operating system.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

The software loaded on a specific workstation is identified and an indication is given of the use of each application.

ASSESSMENT CRITERION 2

Ability to navigate the operating system is demonstrated for specific tasks.

ASSESSMENT CRITERION 3

Simple default settings on a system are customised to suit individual needs with reference to company policy.

ASSESSMENT CRITERION 4

The basic structure of an electronic filing system is explained with reference to drive, directory, folders, files and file types.

ASSESSMENT CRITERION 5

Techniques for managing files in an operating system are demonstrated including, but not limited to, creating folders and copying, moving, deleting, renaming and sorting files and folders.

ASSESSMENT CRITERION 6

The help function is demonstrated for three different tasks.

SPECIFIC OUTCOME 4

Demonstrate basic maintenance procedures.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

Basic maintenance functions are demonstrated for a specific operating system and an indication is given of when to ask for help.

ASSESSMENT CRITERION 2

Procedures to follow when a system stops working are explained with examples.

ASSESSMENT CRITERION 3

The logging off process is demonstrated correctly.

ASSESSMENT CRITERION 4

Procedures to be followed if a virus is suspected are explained with reference to a specific organisation's policies.

UNIT STANDARD ACCREDITATION AND MODERATION OPTIONS

This Unit Standard will be internally assessed by the provider and moderated by a moderator registered by a relevant accredited ETQA or an ETQA that has a Memorandum of Understanding with the relevant accredited ETQA.

UNIT STANDARD ESSENTIAL EMBEDDED KNOWLEDGE

N/A

UNIT STANDARD DEVELOPMENTAL OUTCOME

N/A

UNIT STANDARD LINKAGES

N/A

Critical Cross-field Outcomes (CCFO):**UNIT STANDARD CCFO SCIENCE**

The learner can use technology effectively by operating a computer workstation in a business environment.